

Computational models in Quantum Cryptography

PhD defense

Tristan Nemoz
Supervised by Romain Alléaume and Peter Brown
November 9, 2026



What is cryptography?



Alice

“Meet at 9”

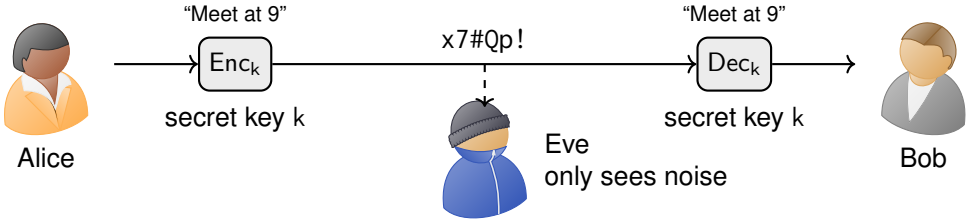


Eve
reads everything

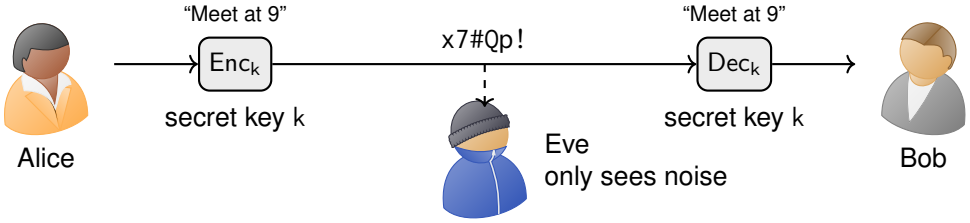


Bob

What is cryptography?

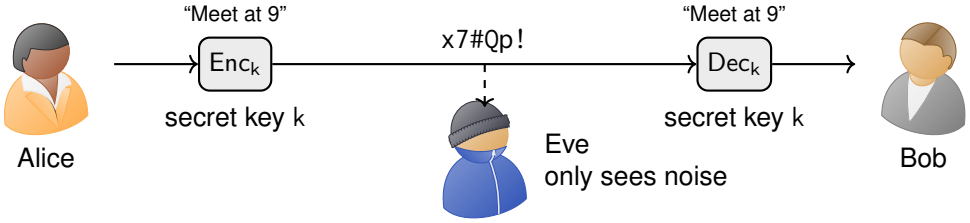


What is cryptography?



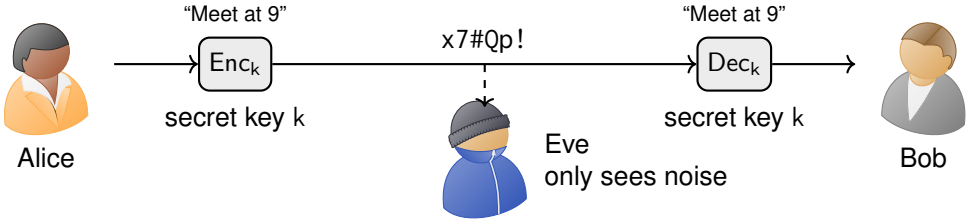
- Goals: confidentiality, integrity, authenticity…

What is cryptography?



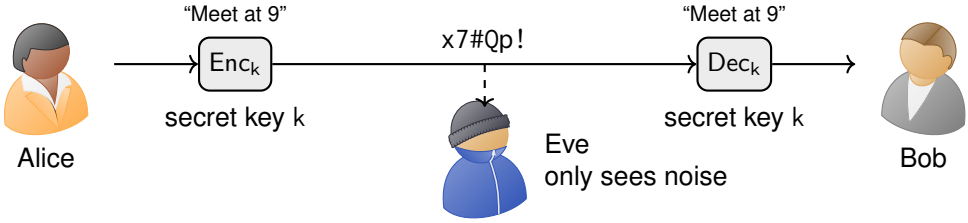
- Goals: confidentiality, integrity, authenticity…
- The protocol is public, only the key k is secret (Kerckhoffs’ principle)

What is cryptography?



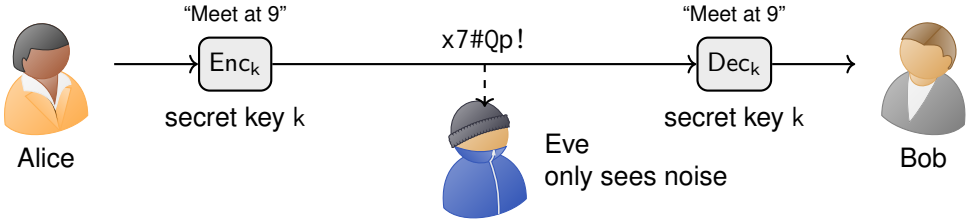
- Goals: confidentiality, integrity, authenticity…
- The protocol is public, only the key k is secret (Kerckhoffs’ principle)
- Security always holds against a model of Eve:

What is cryptography?



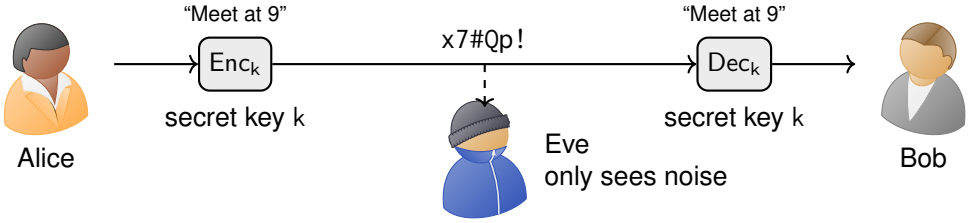
- Goals: confidentiality, integrity, authenticity…
- The protocol is public, only the key k is secret (Kerckhoffs’ principle)
- Security always holds against a model of Eve:
  - *information-theoretic*: Eve has unlimited computing power

What is cryptography?



- Goals: confidentiality, integrity, authenticity…
- The protocol is public, only the key k is secret (Kerckhoffs’ principle)
- Security always holds against a model of Eve:
  - *information-theoretic*: Eve has unlimited computing power
  - *computational*: Eve has a limited computing power and some problem is hard (e.g. factoring)

What is cryptography?



- Goals: confidentiality, integrity, authenticity...
- The protocol is public, only the key k is secret (Kerckhoffs' principle)
- Security always holds against a model of Eve:
  - *information-theoretic*: Eve has unlimited computing power
  - *computational*: Eve has a limited computing power and some problem is hard (e.g. factoring)
- This thesis: explore the limits of quantum constructions, to see where quantum is worth it

A brief history of (post-)quantum cryptography

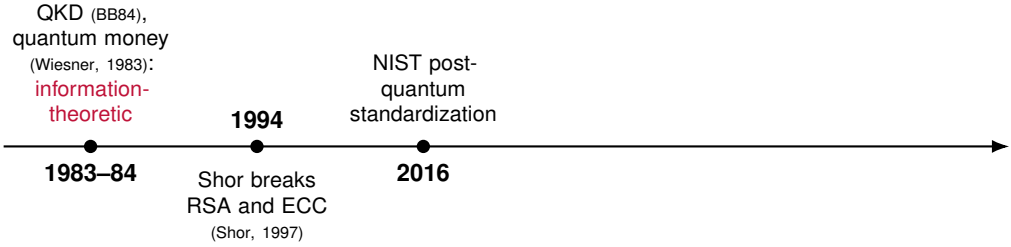
QKD (BB84), quantum money (Wiesner, 1983): information-theoretic

1983–84

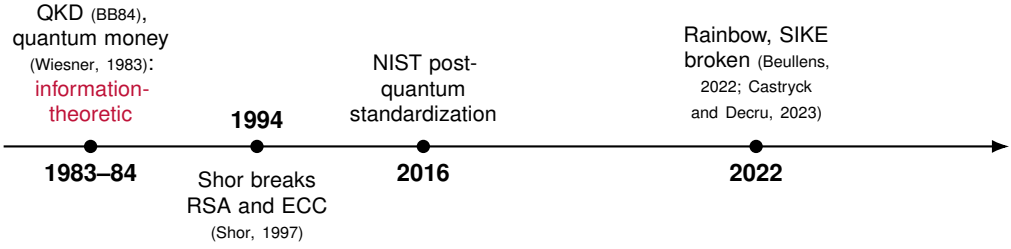
A brief history of (post-)quantum cryptography



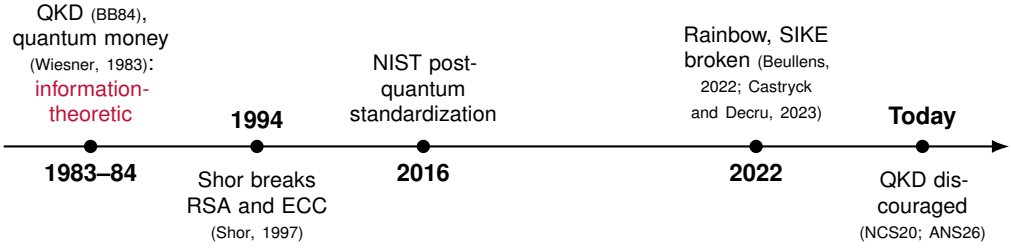
A brief history of (post-)quantum cryptography



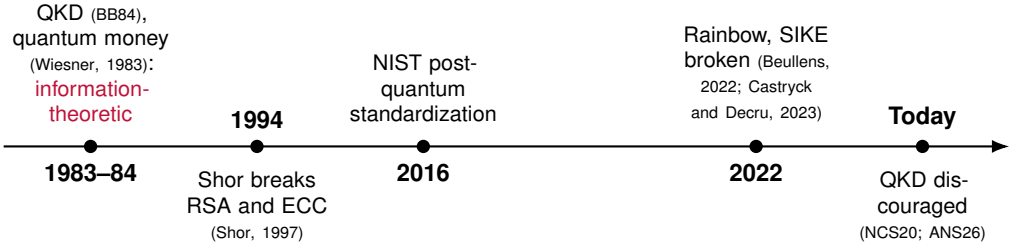
A brief history of (post-)quantum cryptography



A brief history of (post-)quantum cryptography

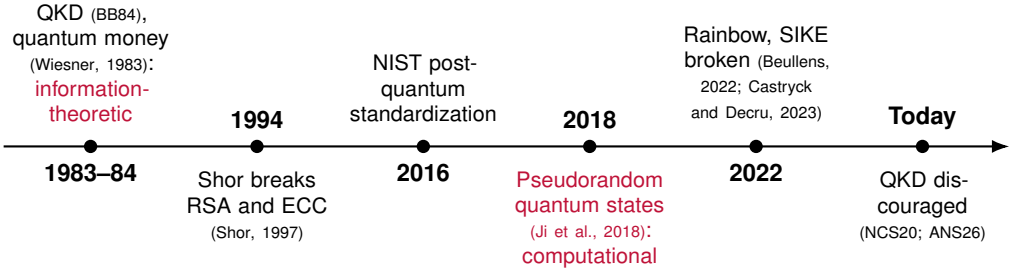


A brief history of (post-)quantum cryptography



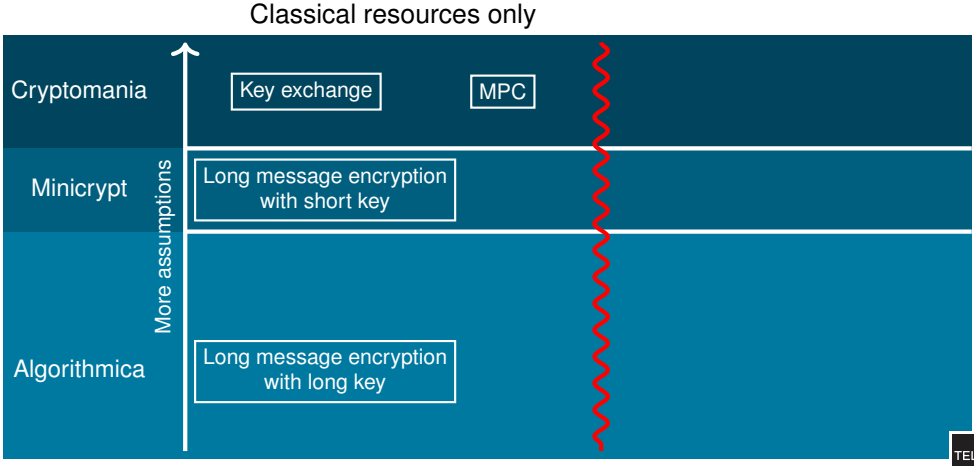
- Information-theoretic security is expensive: dedicated hardware, low rates, short distances

A brief history of (post-)quantum cryptography

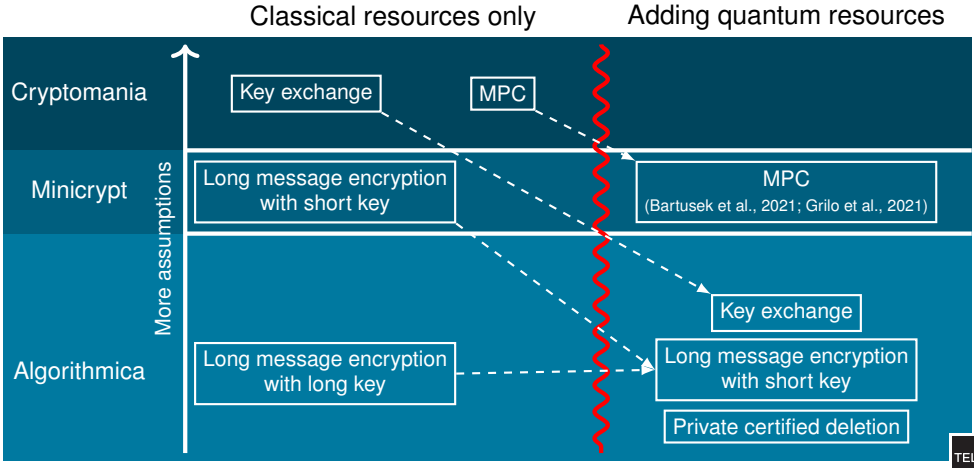


- Information-theoretic security is expensive: dedicated hardware, low rates, short distances
- New trend: accept computational assumptions to try to mitigate these problems

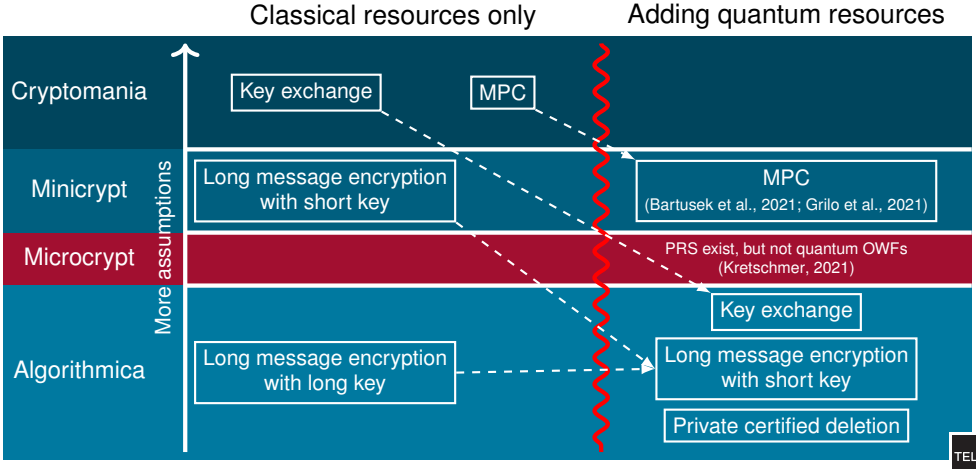
Where can quantum help? Impagliazzo's worlds (Impagliazzo, 1995)



Where can quantum help? Impagliazzo's worlds (Impagliazzo, 1995)



Where can quantum help? Impagliazzo's worlds (Impagliazzo, 1995)



Outline of this talk

Chapter 3

Quantum random number generation

How much secure randomness is in a quantum state?

(Nemoz, Anco and Brown, 2024)

Chapter 4

Real-valued pseudorandom states

Do complex amplitudes increase the security of pseudorandom states?

(Nemoz, Alléaume and Brown, 2026)

Chapter 5

Pseudorandom coherent states

Can pseudorandom states travel on light?

(Kuzaka, Nemoz and Alléaume, in preparation)

Outline of this talk

Chapter 3

Quantum random number generation

How much secure randomness is in a quantum state?

(Nemoz, Anco and Brown, 2024)

Chapter 4

Real-valued pseudorandom states

Do complex amplitudes increase the security of pseudorandom states?

(Nemoz, Alléaume and Brown, 2026)

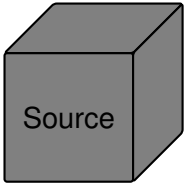
Chapter 5

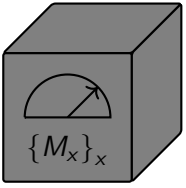
Pseudorandom coherent states

Can pseudorandom states travel on light?

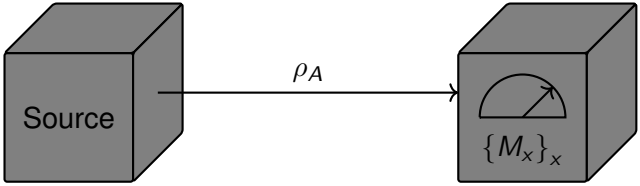
(Kuzaka, Nemoz and Alléaume, in preparation)

Problem setup

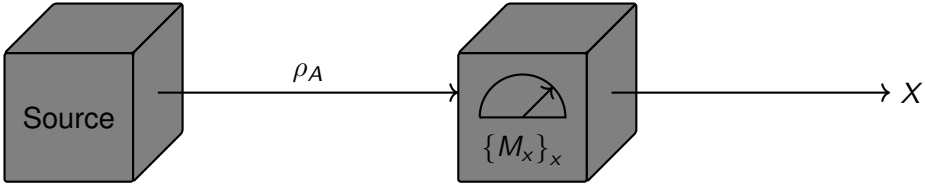




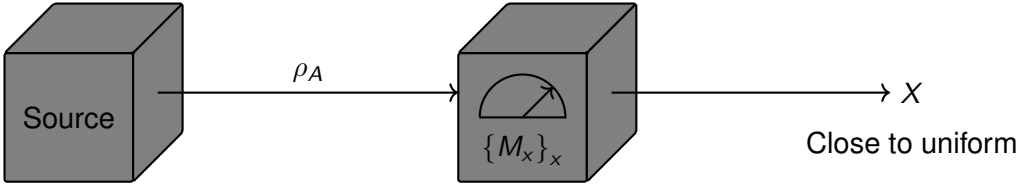
Problem setup



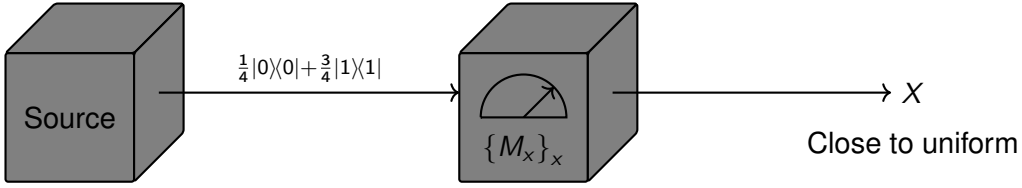
Problem setup



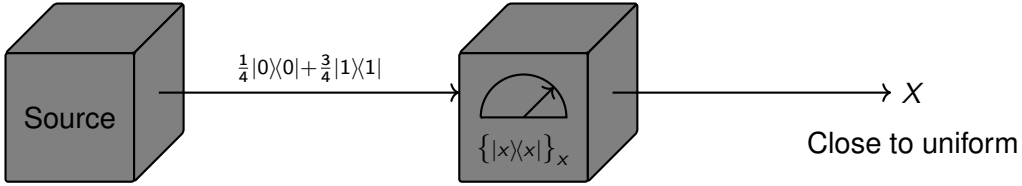
Problem setup



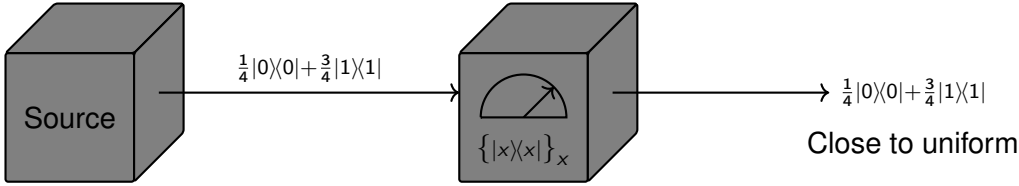
Problem setup



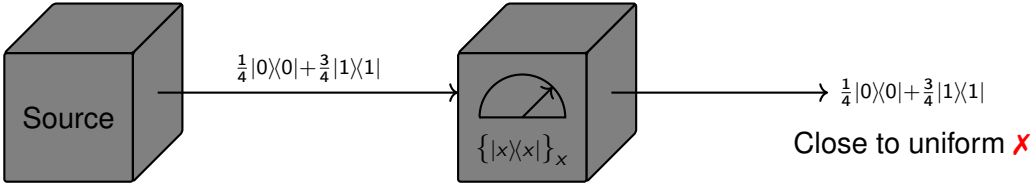
Problem setup



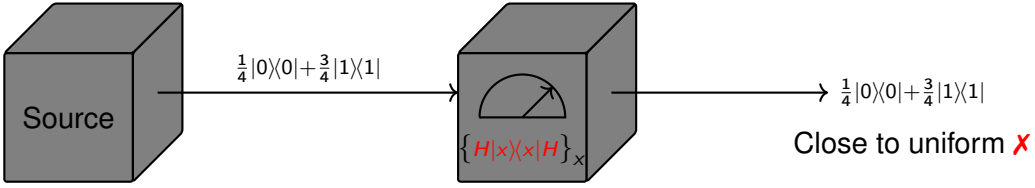
Problem setup



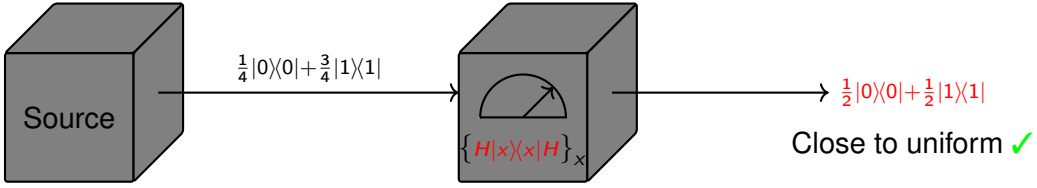
Problem setup



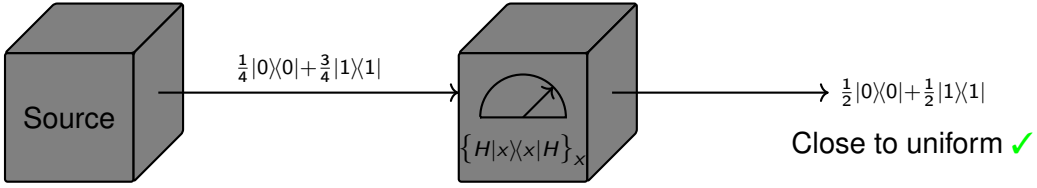
Problem setup



Problem setup

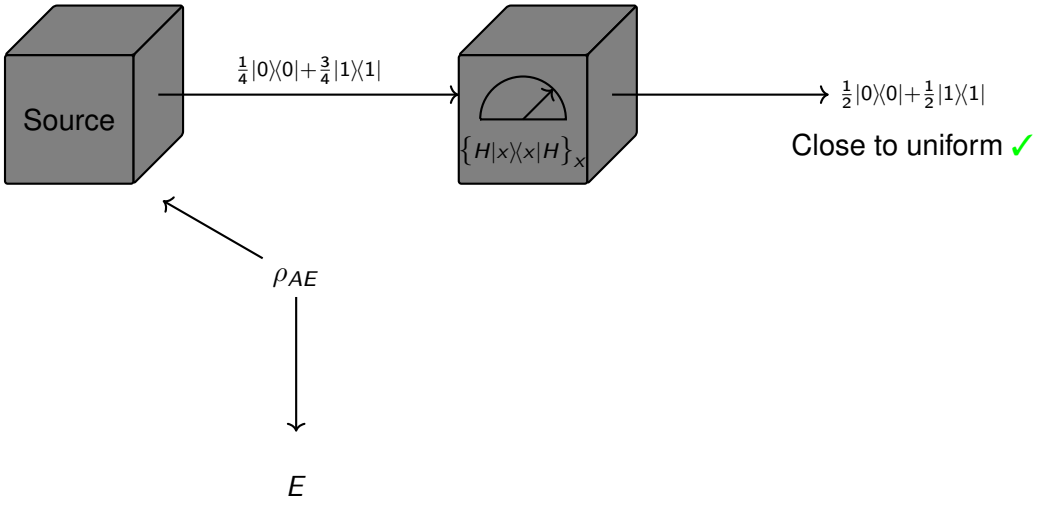


Problem setup

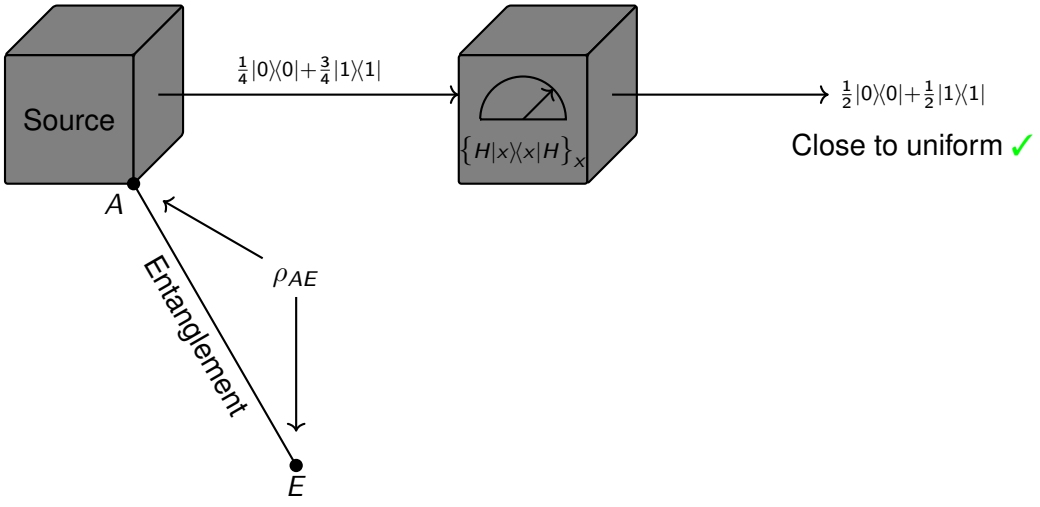


But what if an adversary had information about the source?

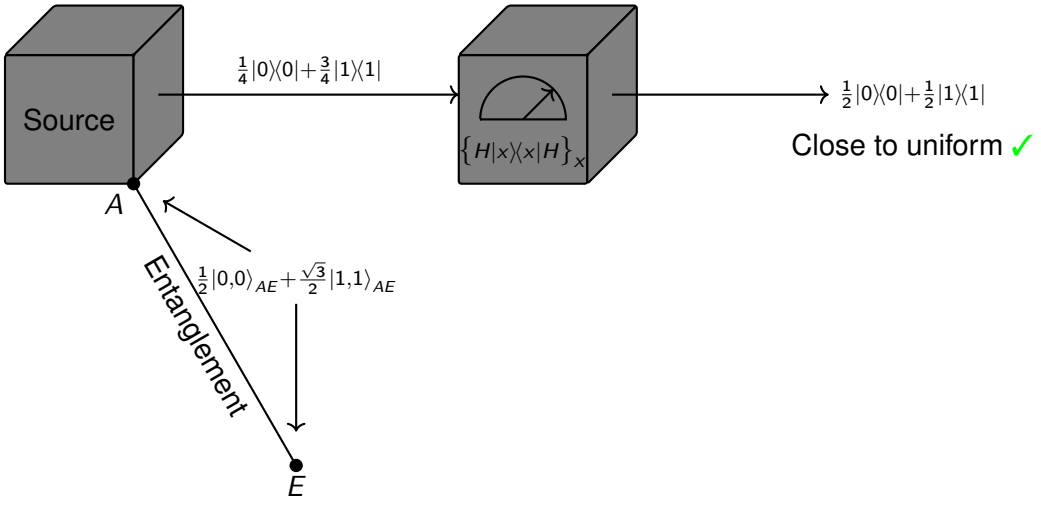
Problem setup



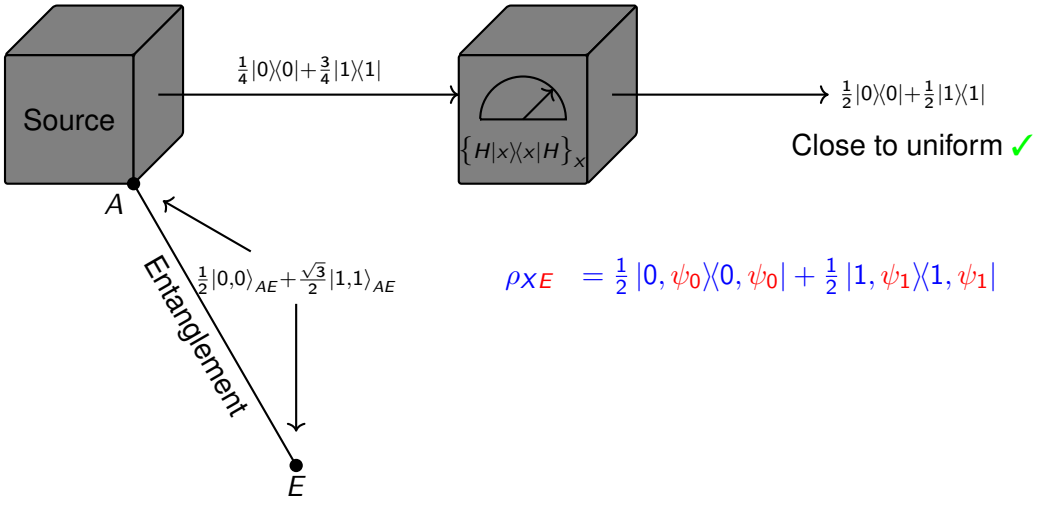
Problem setup



Problem setup

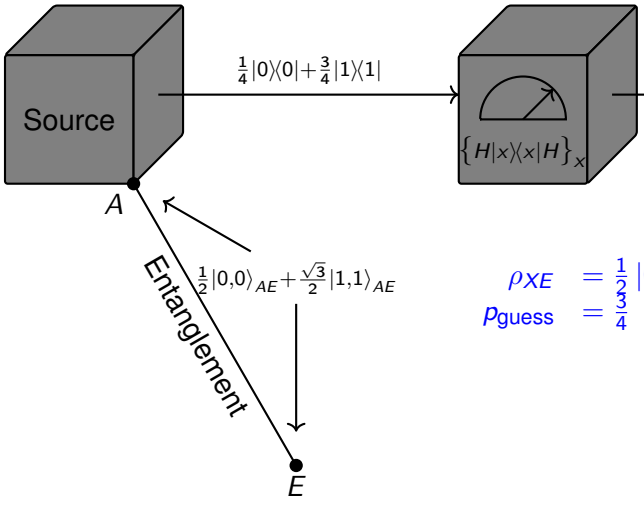


Problem setup



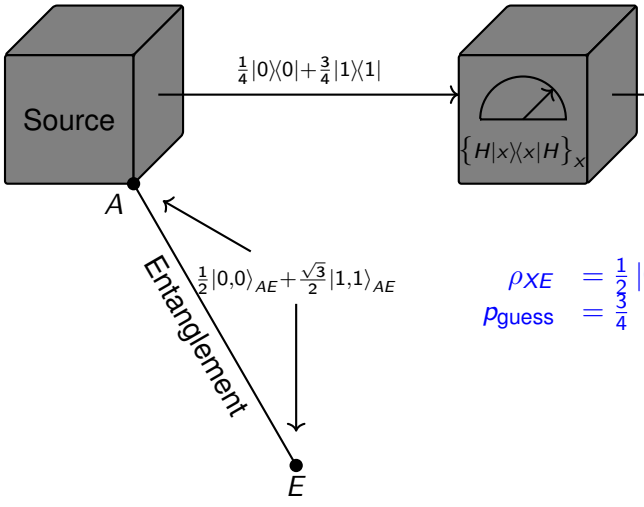
$$\rho_{XE} = \frac{1}{2}|0,\psi_0\rangle\langle 0,\psi_0| + \frac{1}{2}|1,\psi_1\rangle\langle 1,\psi_1|$$

Problem setup



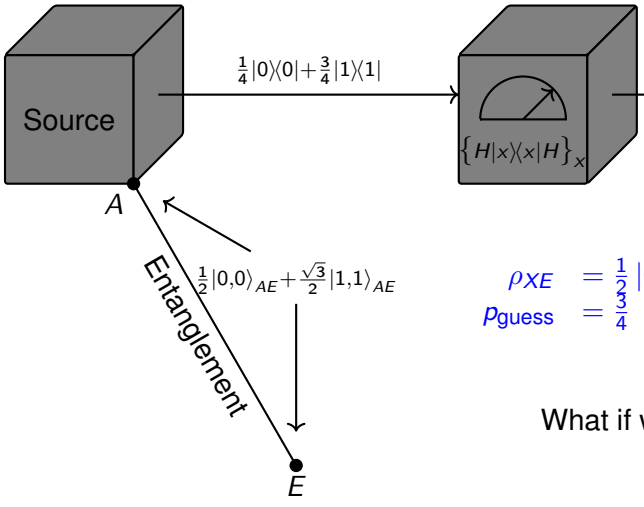
$$\rho_{XE} = \frac{1}{2}|0,\psi_0\rangle\langle 0,\psi_0| + \frac{1}{2}|1,\psi_1\rangle\langle 1,\psi_1|$$
$$p_{\text{guess}} = \frac{3}{4}$$

Problem setup



$$\rho_{XE} = \frac{1}{2}|0,\psi_0\rangle\langle 0,\psi_0| + \frac{1}{2}|1,\psi_1\rangle\langle 1,\psi_1|$$
$$p_{\text{guess}} = \frac{3}{4}$$

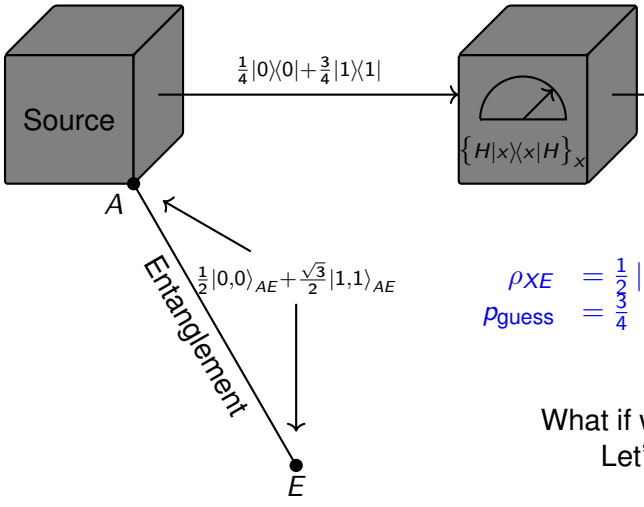
Problem setup



$$\rho_{XE} = \frac{1}{2}|0,\psi_0\rangle\langle 0,\psi_0| + \frac{1}{2}|1,\psi_1\rangle\langle 1,\psi_1|$$
$$p_{\text{guess}} = \frac{3}{4}$$

What if we try to be clever?

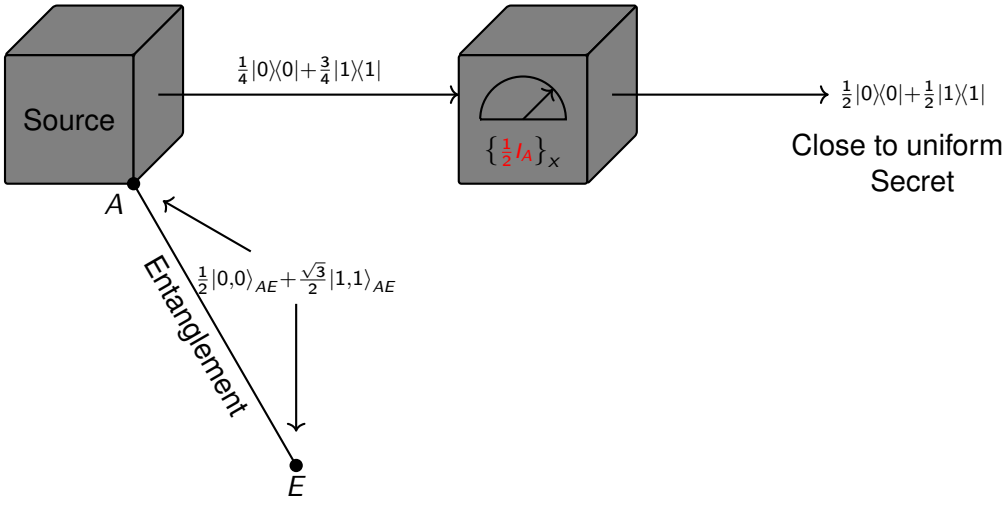
Problem setup



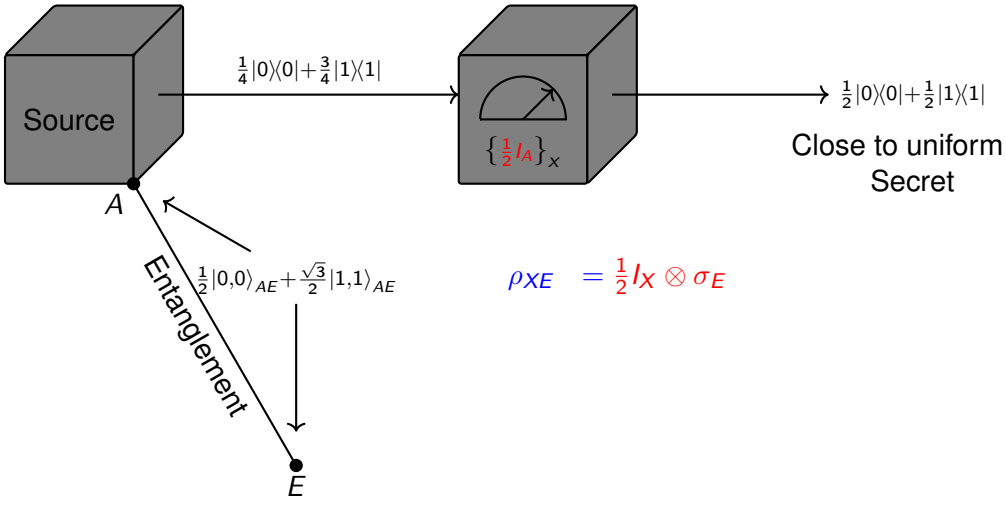
$$\rho_{XE} = \frac{1}{2}|0,\psi_0\rangle\langle 0,\psi_0| + \frac{1}{2}|1,\psi_1\rangle\langle 1,\psi_1|$$
$$p_{\text{guess}} = \frac{3}{4}$$

What if we try to be clever?
Let's use POVMs!

Problem setup

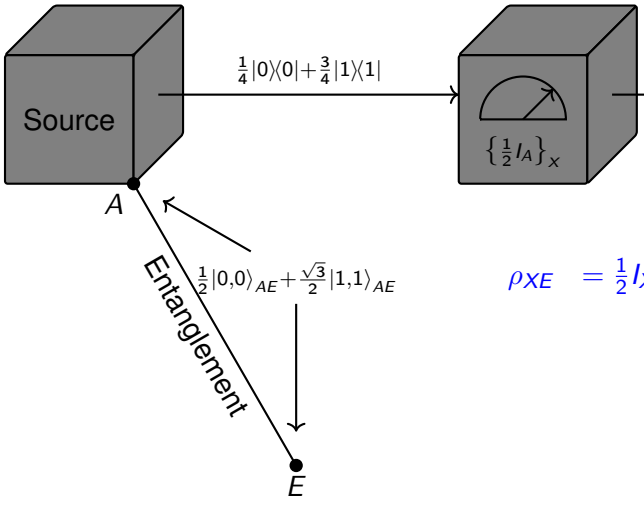


Problem setup



$$\rho_{XE} = \frac{1}{2} I_X \otimes \sigma_E$$

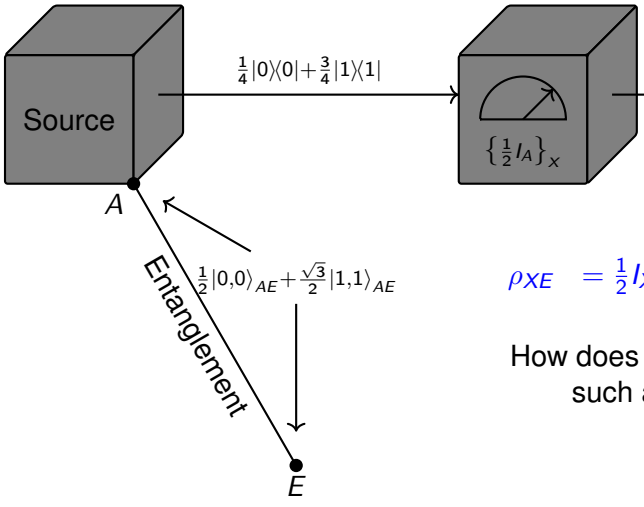
Problem setup



Close to uniform ✓
Secret ✓

$$\rho_{XE} = \frac{1}{2} I_X \otimes \sigma_E$$

Problem setup

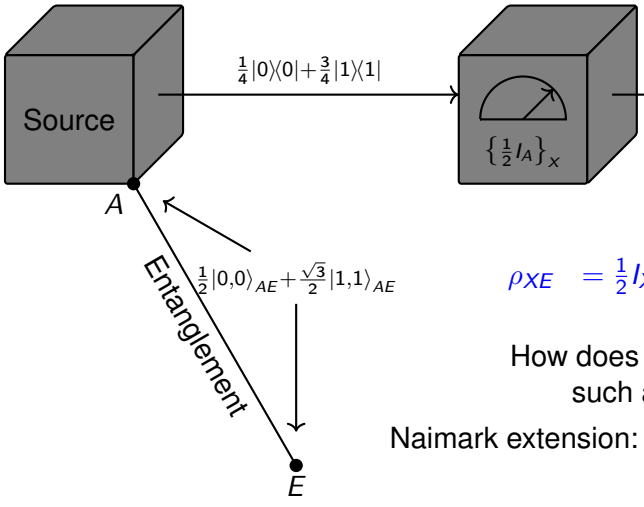


Close to uniform ✓
Secret ✓

$\rho_{XE} = \frac{1}{2} I_X \otimes \sigma_E$

How does the device implement such a measurement?

Problem setup



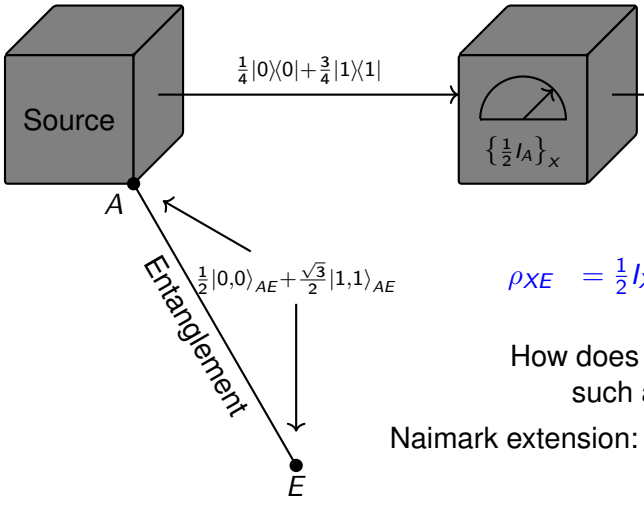
Close to uniform ✓
Secret ✓

$$\rho_{XE} = \frac{1}{2} I_X \otimes \sigma_E$$

How does the device implement such a measurement?

Naimark extension: measures first qubit of $\frac{|00\rangle + |11\rangle}{\sqrt{2}}$

Problem setup

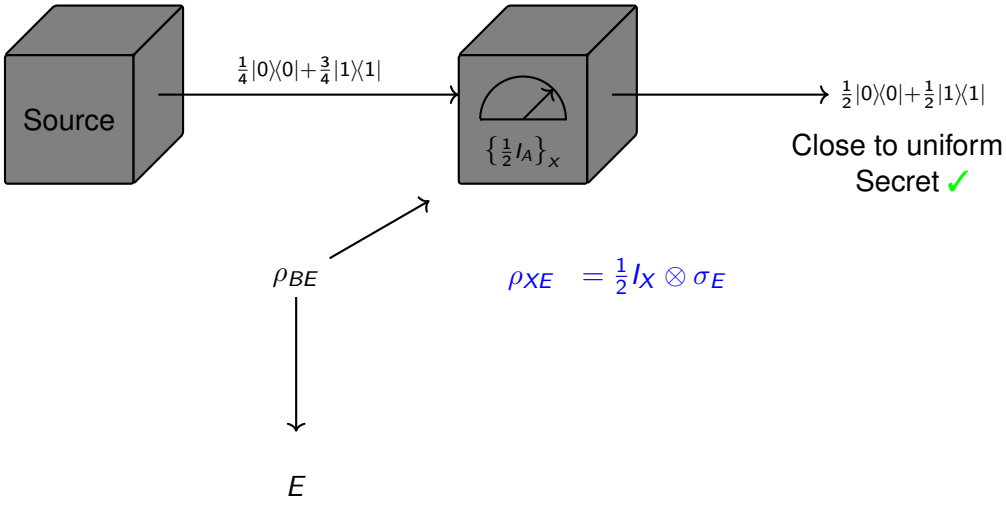


$$\rho_{XE} = \frac{1}{2} I_X \otimes \sigma_E$$

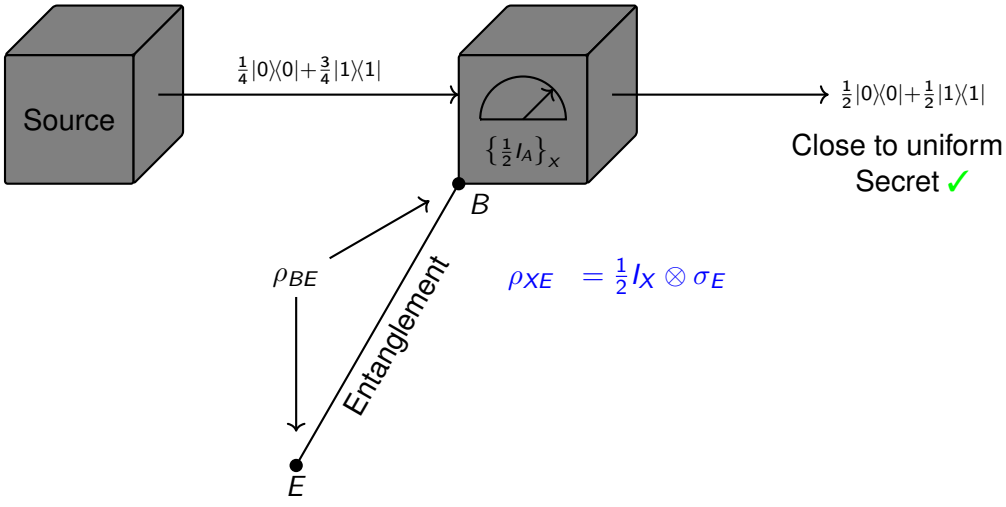
How does the device implement such a measurement?

Naimark extension: measures first qubit of $\frac{|00\rangle + |11\rangle}{\sqrt{2}}$

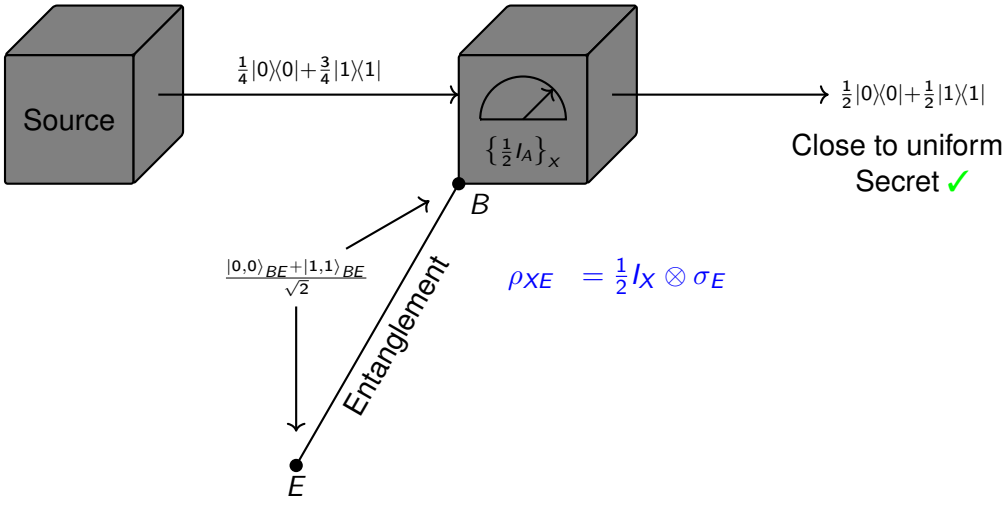
Problem setup



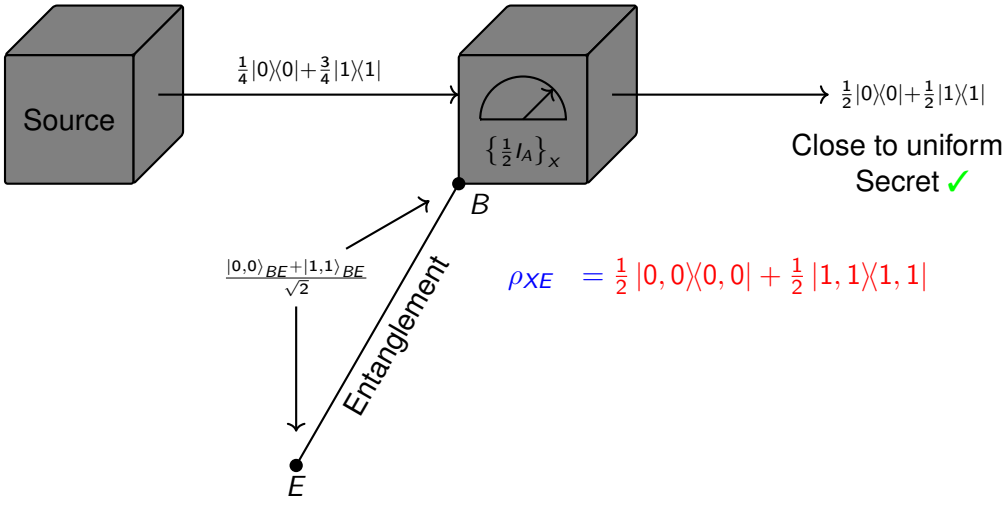
Problem setup



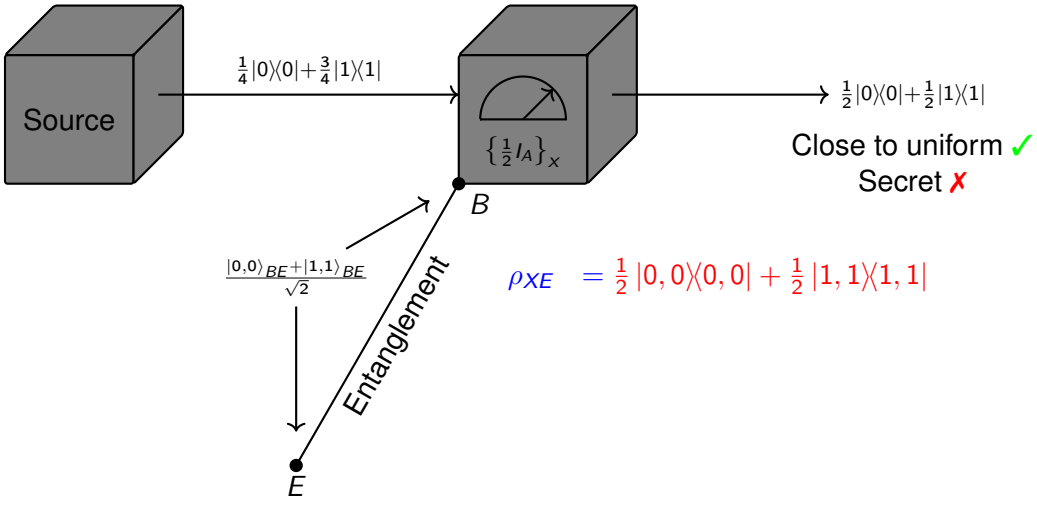
Problem setup



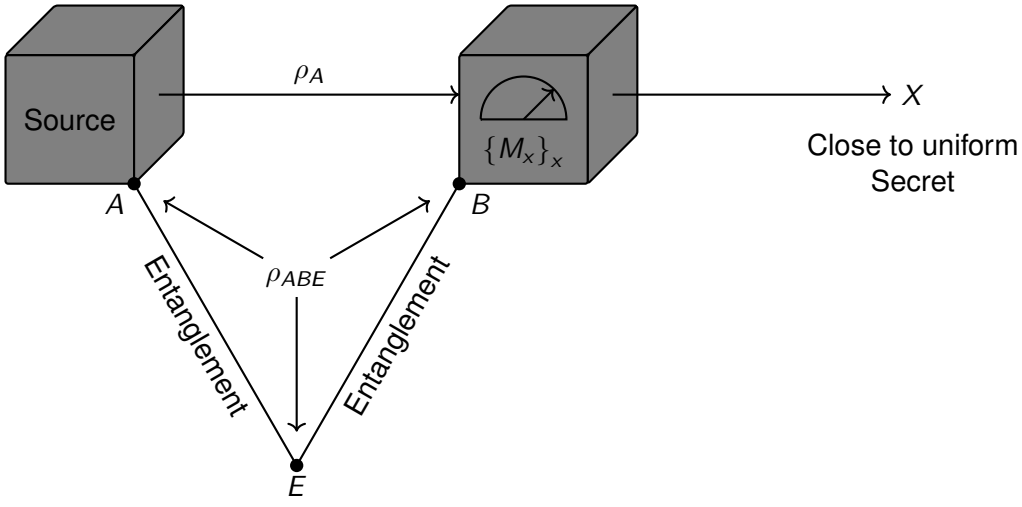
Problem setup



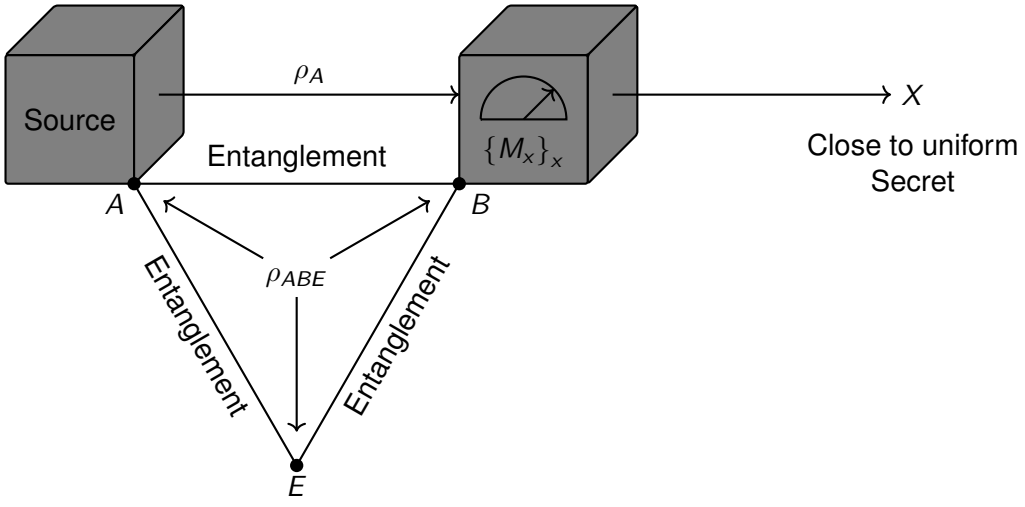
Problem setup



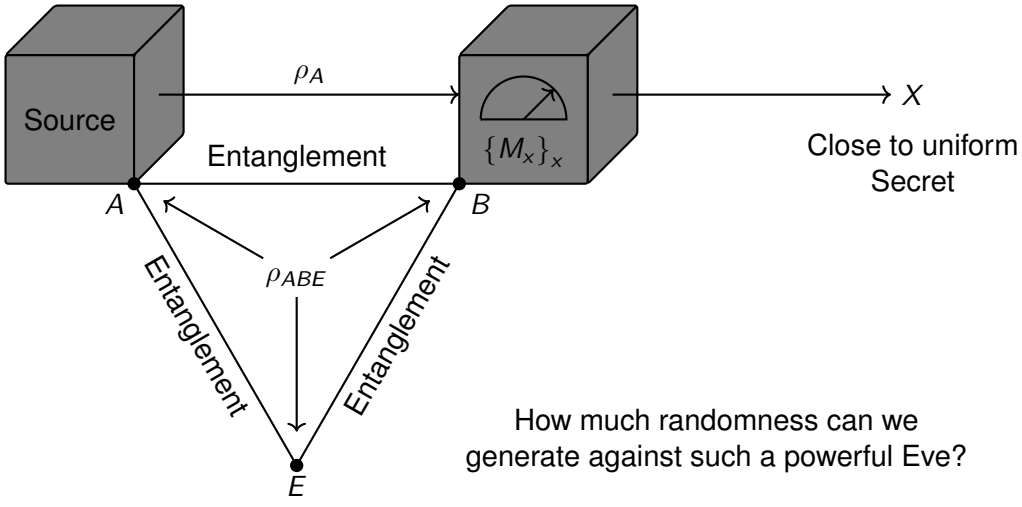
Problem setup



Problem setup

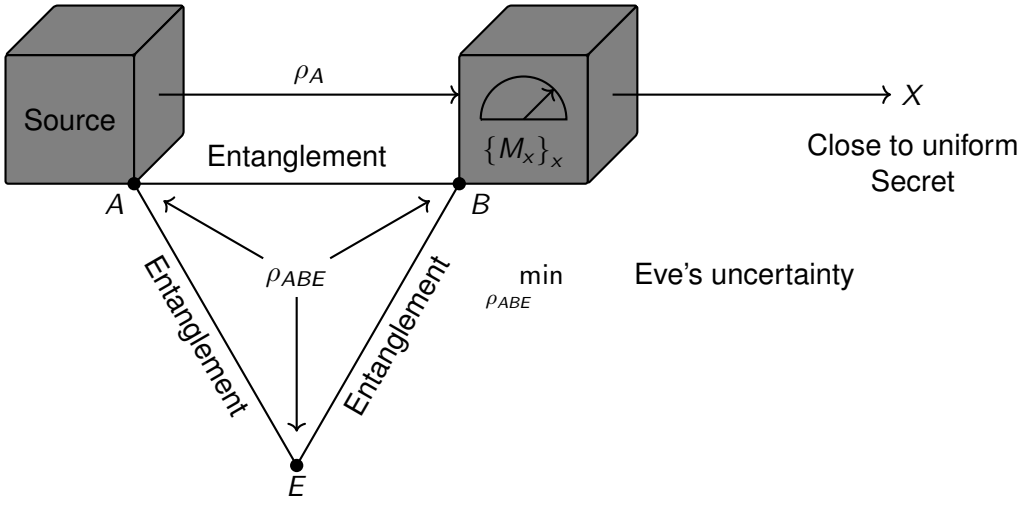


Problem setup

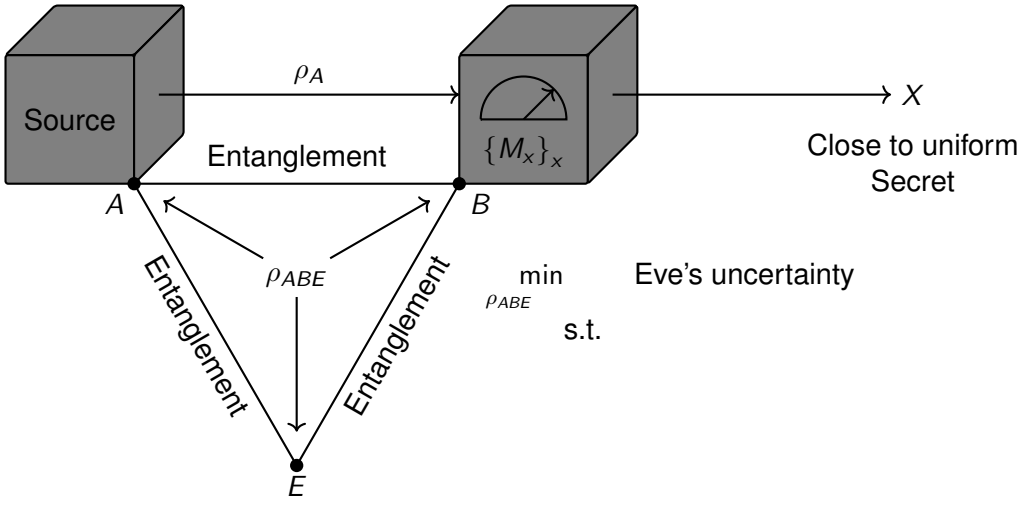


How much randomness can we generate against such a powerful Eve?

Problem setup



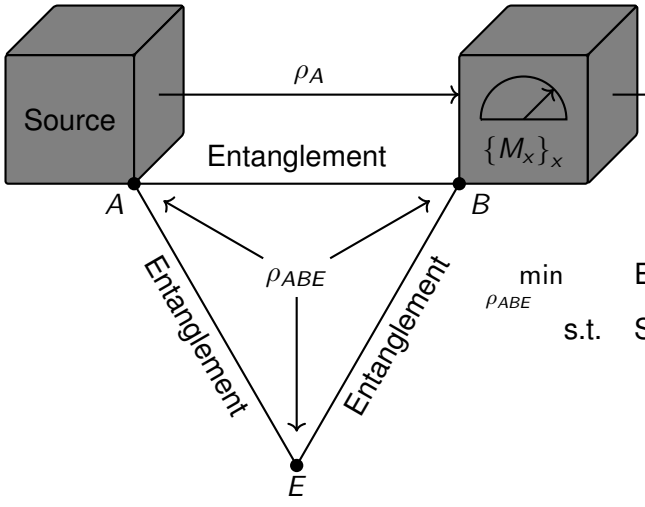
Problem setup



$$\min_{\rho_{ABE}} \quad \text{Eve's uncertainty}$$

$$\text{s.t.}$$

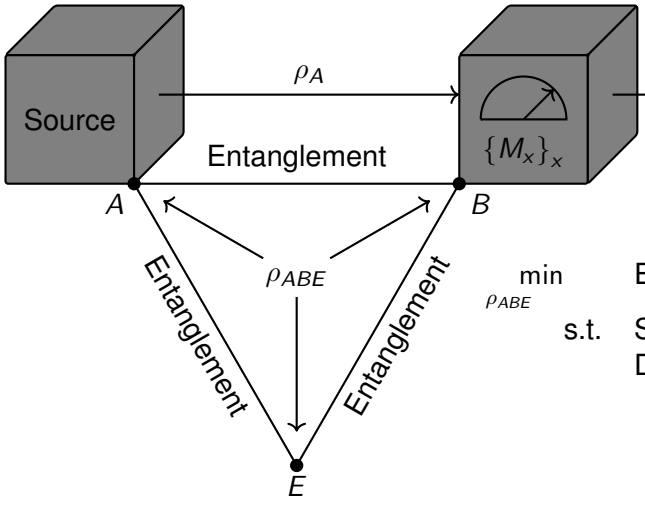
Problem setup



$$\min_{\rho_{ABE}} \quad \text{Eve's uncertainty}$$

$$\text{s.t.} \quad \text{Source produces } \rho_A$$

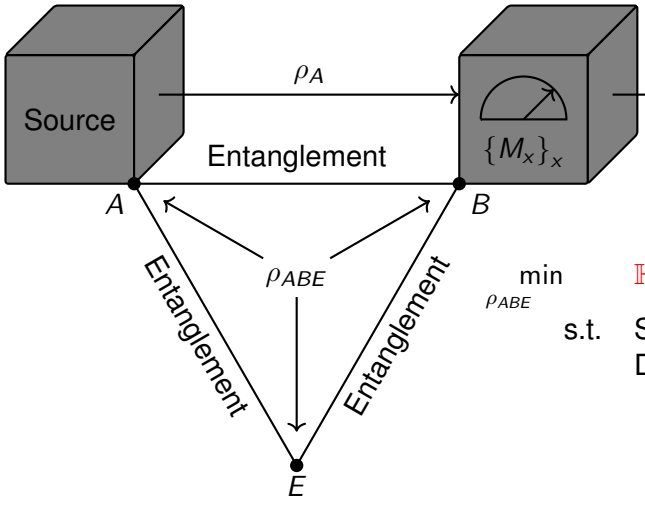
Problem setup



$$\min_{\rho_{ABE}} \quad \text{Eve's uncertainty}$$

s.t. Source produces ρ_A
Device implements $\{M_x\}_x$

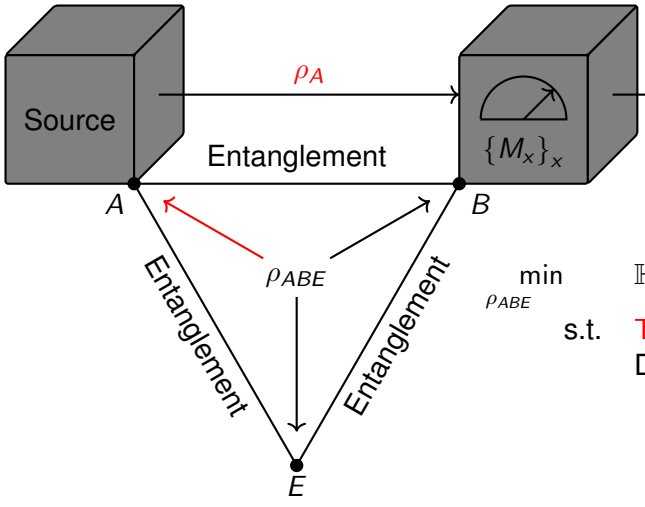
Problem setup



$$\min_{\rho_{ABE}} \quad \mathbb{H}(X|E)_{\rho_{XE}}$$

s.t. Source produces ρ_A
Device implements $\{M_x\}_x$

Problem setup

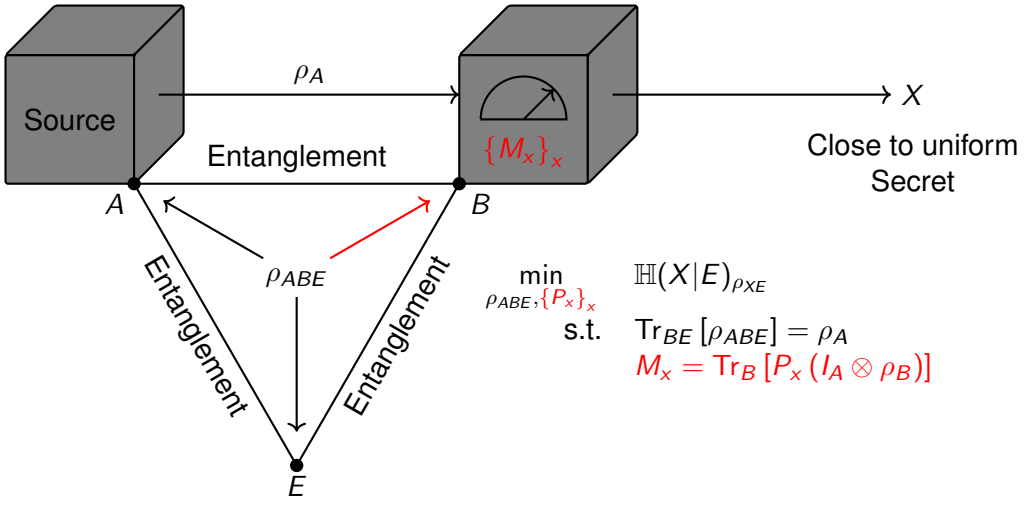


$$\min_{\rho_{ABE}} \quad \mathbb{H}(X|E)_{\rho_{XE}}$$

s.t. $\mathrm{Tr}_{BE}[\rho_{ABE}] = \rho_A$

Device implements $\{M_x\}_x$

Problem setup

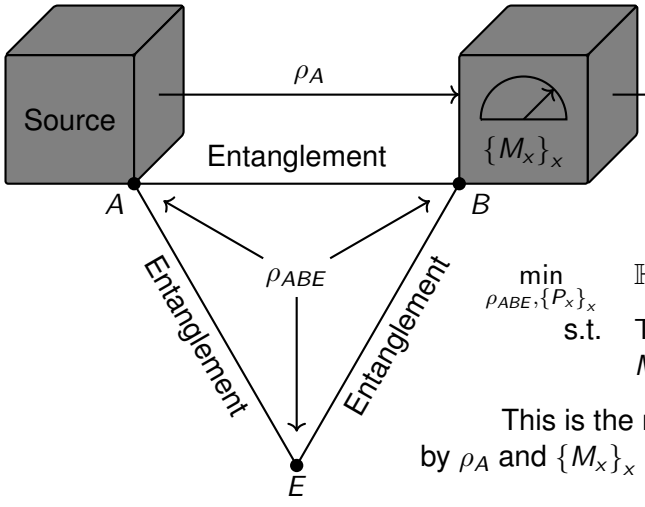


$$\min_{\rho_{ABE},\{P_x\}_x} \quad \mathbb{H}(X|E)_{\rho_{XE}}$$

$$\text{s.t.} \quad \mathrm{Tr}_{BE}\left[\rho_{ABE}\right] = \rho_A$$

$$M_x = \mathrm{Tr}_B\left[P_x\left(I_A \otimes \rho_B\right)\right]$$

Problem setup



$$\min_{\rho_{ABE},\{P_x\}_x} \quad \mathbb{H}(X|E)_{\rho_{XE}}$$
$$\text{s.t.} \quad \mathrm{Tr}_{BE}\left[\rho_{ABE}\right] = \rho_A$$
$$M_x = \mathrm{Tr}_B\left[P_x\left(I_A \otimes \rho_B\right)\right]$$

This is the randomness produced
by ρ_A and $\{M_x\}_x$ against a general adversary!

Problem statement

- Randomness hand-waved using entropies

Problem statement

- Randomness hand-waved using entropies
- ε-secure randomness:

$$\rho_{XE} \approx_\varepsilon \frac{1}{d_X} I_X \otimes \rho_E$$

Problem statement

- Randomness hand-waved using entropies
- ε-secure randomness:

$$\rho_{XE} \approx_\varepsilon \frac{1}{d_X} I_X \otimes \rho_E$$

  - Uniformly random

Problem statement

- Randomness hand-waved using entropies
- ε-secure randomness:

$$\rho_{XE} \approx_\varepsilon \frac{1}{d_X} I_X \otimes \rho_E$$

  - Uniformly random, independent of Eve

Problem statement

- Randomness hand-waved using entropies
- ε-secure randomness:

$$\rho_{XE} \approx_\varepsilon \frac{1}{d_X} I_X \otimes \rho_E$$

  - Uniformly random, independent of Eve
  - (Dupuis, 2023): link with sandwiched Rényi entropies H_α

Problem statement

- Randomness hand-waved using entropies
- ε-secure randomness:

$$\rho_{XE} \approx_\varepsilon \frac{1}{d_X} I_X \otimes \rho_E$$

  - Uniformly random, independent of Eve
  - (Dupuis, 2023): link with sandwiched Rényi entropies H_α

$$\min_{\rho_{AEB}, \{P_x\}_x} \mathbb{H}(X|E)_{\rho_{XE}} = \quad \text{randomness produced by } \rho_A \text{ and } \{M_x\}_x$$

Problem statement

- Randomness hand-waved using entropies
- ε-secure randomness:

$$\rho_{XE} \approx_\varepsilon \frac{1}{d_X} I_X \otimes \rho_E$$

  - Uniformly random, independent of Eve
  - (Dupuis, 2023): link with sandwiched Rényi entropies H_α

$$\sup_{\{M_x\}_x} \min_{\rho_{AEB}, \{P_x\}_x} \mathbb{H}(X|E)_{\rho_{XE}} = \text{maximal randomness produced by } \rho_A$$

Problem statement

- Randomness hand-waved using entropies
- ε-secure randomness:

$$\rho_{XE} \approx_\varepsilon \frac{1}{d_X} I_X \otimes \rho_E$$

  - Uniformly random, independent of Eve
  - (Dupuis, 2023): link with sandwiched Rényi entropies H_α

$$\sup_{\{M_x\}_x} \min_{\rho_{AEB}, \{P_x\}_x} \mathbb{H}(X|E)_{\rho_{XE}} = \text{ maximal randomness produced by } \rho_A$$

- "max min" problem

Problem statement

- Randomness hand-waved using entropies
- ε-secure randomness:

$$\rho_{XE} \approx_\varepsilon \frac{1}{d_X} I_X \otimes \rho_E$$

  - Uniformly random, independent of Eve
  - (Dupuis, 2023): link with sandwiched Rényi entropies H_α

$$\sup_{\{M_x\}_x} \min_{\rho_{AEB}, \{P_x\}_x} \mathbb{H}(X|E)_{\rho_{XE}} = \text{ maximal randomness produced by } \rho_A$$

- "max min" problem
- $\dim(E)$ arbitrary

Problem statement

- Randomness hand-waved using entropies
- ε-secure randomness:

$$\rho_{XE} \approx_\varepsilon \frac{1}{d_X} I_X \otimes \rho_E$$

  - Uniformly random, independent of Eve
  - (Dupuis, 2023): link with sandwiched Rényi entropies H_α

$$\sup_{\{M_x\}_x} \min_{\rho_{AEB}, \{P_x\}_x} \mathbb{H}(X|E)_{\rho_{XE}} = \text{ maximal randomness produced by } \rho_A$$

- "max min" problem
- $\dim(E)$ arbitrary

- Optimize over all Naimark extensions

Problem statement

- Randomness hand-waved using entropies
- ε-secure randomness:

$$\rho_{XE} \approx_\varepsilon \frac{1}{d_X} I_X \otimes \rho_E$$

  - Uniformly random, independent of Eve
  - (Dupuis, 2023): link with sandwiched Rényi entropies H_α

$$\sup_{\{M_x\}_x} \min_{\rho_{AEB}, \{P_x\}_x} \mathbb{H}(X|E)_{\rho_{XE}} = \text{ maximal randomness produced by } \rho_A$$

- "max min" problem
- $\dim(E)$ arbitrary

- Optimize over all Naimark extensions
  - $\dim(B)$ arbitrary

Problem statement

- Randomness hand-waved using entropies
- ε-secure randomness:

$$\rho_{XE} \approx_\varepsilon \frac{1}{d_X} I_X \otimes \rho_E$$

  - Uniformly random, independent of Eve
  - (Dupuis, 2023): link with sandwiched Rényi entropies H_α

$$\sup_{\{M_x\}_x} \min_{\rho_{AEB}, \{P_x\}_x} \mathbb{H}(X|E)_{\rho_{XE}} = \text{ maximal randomness produced by } \rho_A$$

- "max min" problem
- $\dim(E)$ arbitrary

- Optimize over all Naimark extensions
  - $\dim(B)$ arbitrary
- $\implies$ Not trivial!

Reductions

$$\sup_{\text{POVM } \{M_x\}_x} \min_{\substack{\rho_{ABE} \\ \text{PVM } \{P_x\}_x}} \mathbb{H}(X|E)_{\rho_{XE}}$$

$$\text{s.t.} \quad \begin{aligned} &\mathrm{Tr}_{BE}\left[\rho_{ABE}\right] = \rho_A \\ &M_x = \mathrm{Tr}_B\left[P_x \left(I_A \otimes \mathrm{Tr}_{AE}\left[\rho_{ABE}\right]\right)\right] \end{aligned}$$

Reductions

$$\sup_{\text{POVM } \{M_x\}_x} \min_{\substack{\rho_{ABE} \\ \text{PVM } \{P_x\}_x}} \mathbb{H}(X|E)_{\rho_{XE}}$$

$$\text{s.t.} \quad \mathrm{Tr}_{BE}\left[\rho_{ABE}\right] = \rho_A$$
$$M_x = \mathrm{Tr}_B\left[P_x\left(I_A \otimes \mathrm{Tr}_{AE}\left[\rho_{ABE}\right]\right)\right]$$

- Eve: purifies ρ_{AB}

Reductions

$$\sup_{\text{POVM } \{M_x\}_x} \min_{\text{PVM } \{P_x\}_x} \mathbb{H}(X|E)_{\rho_{XE}}$$

$$\text{s.t.} \quad \mathrm{Tr}_{BE}\left[|\psi_{ABE}\rangle\langle\psi_{ABE}|\right] = \rho_A$$
$$M_x = \mathrm{Tr}_B\left[P_x\left(I_A \otimes \mathrm{Tr}_{AE}\left[|\psi_{ABE}\rangle\langle\psi_{ABE}|\right]\right)\right]$$

- Eve: purifies ρ_{AB}

Reductions

$$\sup_{\text{POVM } \{M_x\}_x} \min_{\text{PVM } \{P_x\}_x} \mathbb{H}(X|E)_{\rho_{XE}}$$
$$\text{s.t.} \quad \mathrm{Tr}_{BE}\left[|\psi_{ABE}\rangle\langle\psi_{ABE}|\right] = \rho_A$$
$$M_x = \mathrm{Tr}_B\left[P_x\left(I_A \otimes \mathrm{Tr}_{AE}\left[|\psi_{ABE}\rangle\langle\psi_{ABE}|\right]\right)\right]$$

- Eve: purifies ρ_{AB}
- Alice: picks rank-1 extremal POVM

Reductions

$$\sup_{\text{Extremal rank-1 POVM } \{M_x\}_x} \min_{\text{PVM } \{P_x\}_x} \mathbb{H}(X|E)_{\rho_{XE}}$$

$$\text{s.t.} \quad \mathrm{Tr}_{BE}\left[|\psi_{ABE}\rangle\langle\psi_{ABE}|\right] = \rho_A$$
$$M_x = \mathrm{Tr}_B\left[P_x\left(I_A \otimes \mathrm{Tr}_{AE}\left[|\psi_{ABE}\rangle\langle\psi_{ABE}|\right]\right)\right]$$

- Eve: purifies ρ_{AB}
- Alice: picks rank-1 extremal POVM

Reductions

$$\sup_{\text{Extremal rank-1 POVM } \{M_x\}_x} \min_{\text{PVM } \{P_x\}_x} \mathbb{H}(X|E)_{\rho_{XE}}$$

$$\text{s.t.} \quad \mathrm{Tr}_{BE}\left[|\psi_{ABE}\rangle\langle\psi_{ABE}|\right] = \rho_A$$
$$M_x = \mathrm{Tr}_B\left[P_x\left(I_A \otimes \mathrm{Tr}_{AE}\left[|\psi_{ABE}\rangle\langle\psi_{ABE}|\right]\right)\right]$$

- Eve: purifies ρ_{AB}
- Alice: picks rank-1 extremal POVM
- Technical result: M rank-1 extremal $\implies \rho_{XE} \perp\!\!\!\perp \rho_B$

Reductions

$$\sup_{\text{Extremal rank-1 POVM } \{M_x\}_x} \mathbb{H}(X|E)_{\rho_{XE}}$$

$$\text{s.t.} \quad \mathrm{Tr}_E\left[|\psi_{AE}\rangle\langle\psi_{AE}|\right] = \rho_A$$

- Eve: purifies ρ_{AB}
- Alice: picks rank-1 extremal POVM
- Technical result: M rank-1 extremal $\implies \rho_{XE} \perp\!\!\!\perp \rho_B$

Reductions

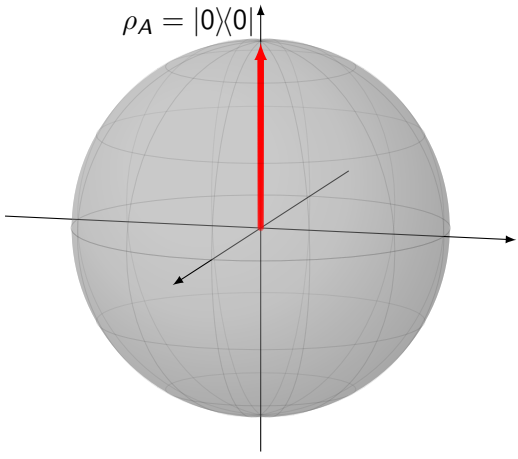
$$\sup_{\text{Extremal rank-1 POVM } \{M_x\}_x} \mathbb{H}(X|E)_{\rho_{XE}}$$

$$\text{s.t.} \quad \mathrm{Tr}_E\left[|\psi_{AE}\rangle\langle\psi_{AE}|\right] = \rho_A$$

- Eve: purifies ρ_{AB}
- Alice: picks rank-1 extremal POVM
- Technical result: M rank-1 extremal $\implies \rho_{XE} \perp\!\!\!\perp \rho_B$

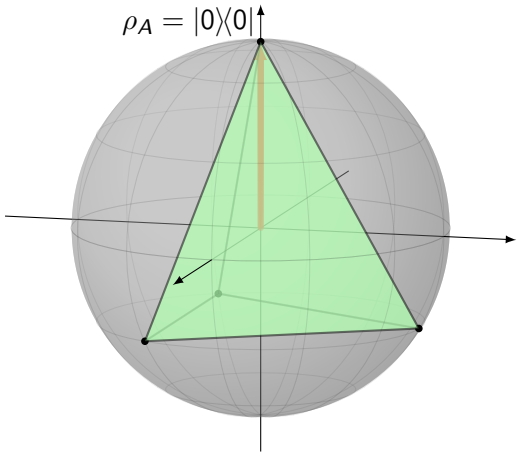
$\implies$ We only need to find the best rank-1 extremal POVM!

A simple case

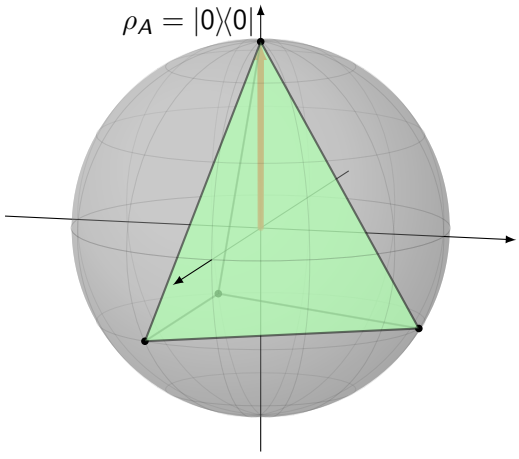


A simple case

Extremal

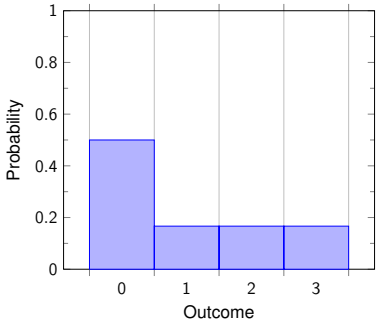


A simple case

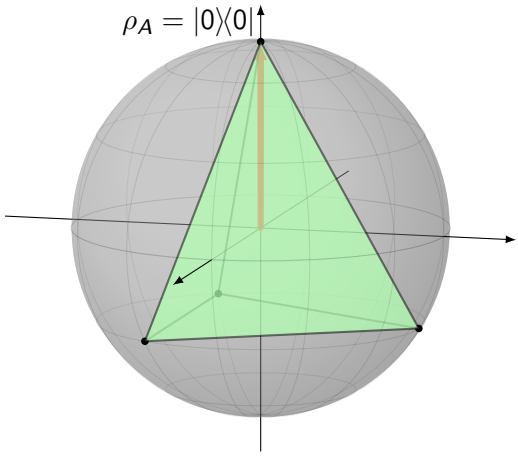


Extremal

$\min \mathbb{H}(X|E) \approx 1.792$



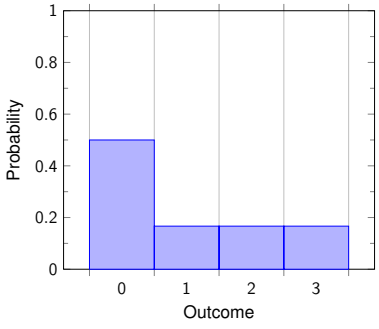
A simple case



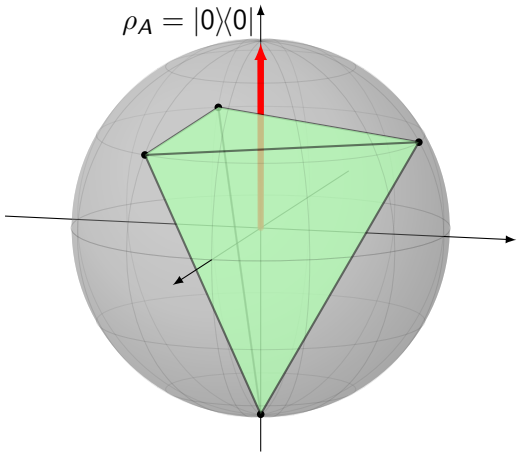
Extremal

$\min \mathbb{H}(X|E) \approx 1.792$

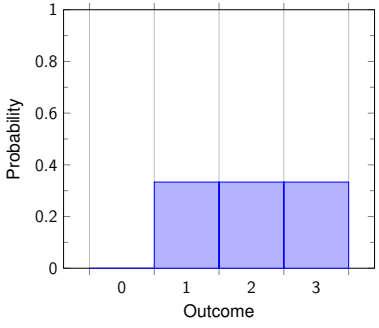
SIC-POVM



A simple case



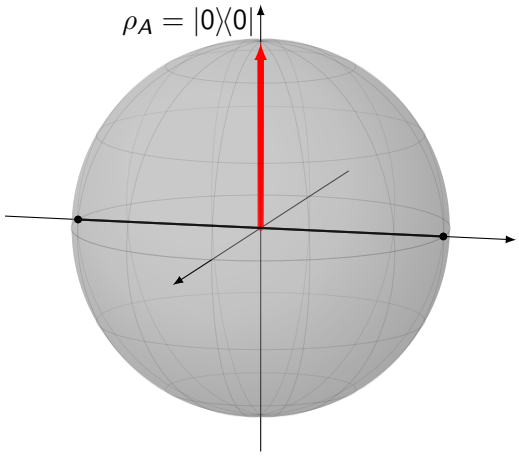
Extremal
$\min \mathbb{H}(X|E) \approx 1.585$
SIC-POVM

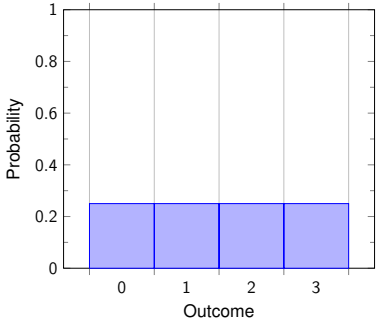


A simple case

Not extremal

$\min \mathbb{H}(X|E) = 1$



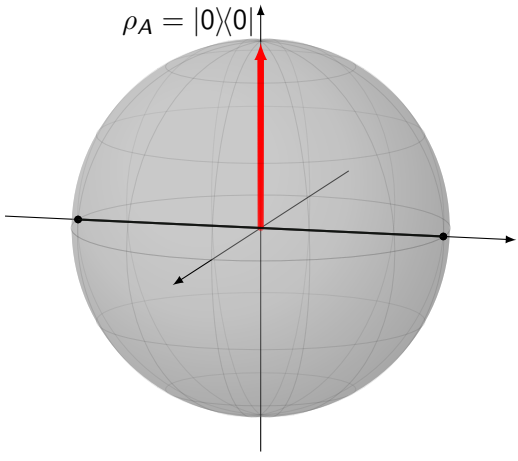


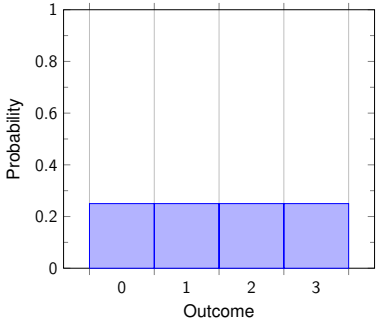
$$M = \left\{ \tfrac{1}{2} |+\rangle\langle +| , \tfrac{1}{2} |+\rangle\langle +| , \tfrac{1}{2} |-\rangle\langle -| , \tfrac{1}{2} |-\rangle\langle -| \right\}$$

A simple case

Not extremal

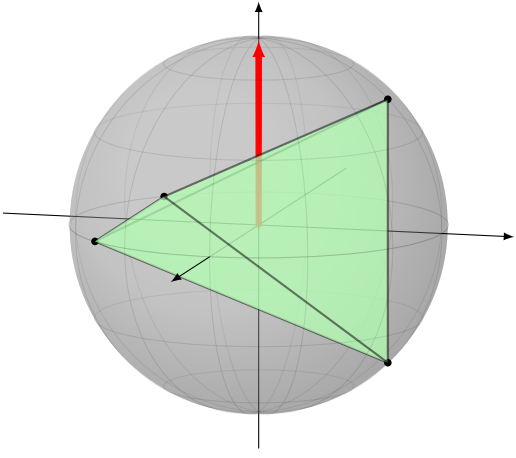
$\min \mathbb{H}(X|E) = 1$





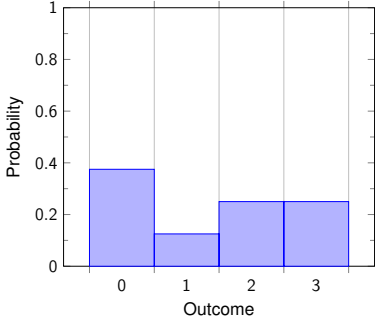
Key result: extremal rank-1 POVMs are dense within rank-1 POVMs

A simple case



Extremal

$\min \mathbb{H}(X|E) \approx 2$



Main result

Theorem

Let ρ_A be a d_A-dimensional quantum state and $\varepsilon > 0$.

Main result

Theorem

Let ρ_A be a d_A-dimensional quantum state and $\varepsilon > 0$. One can extract ℓ bits of ε-secure randomness from $\rho_A^{\otimes n}$ whenever

$$\frac{\ell}{n} \leqslant 2\log\left(d_A\right) - \min_{\alpha\in(1,2]} \left(H_{\frac{\alpha}{2\alpha-1}}(A) - \frac{\alpha}{n(\alpha-1)} \log(\varepsilon) \right) + \frac{2}{n}.$$

Main result

Theorem

Let ρ_A be a d_A-dimensional quantum state and $\varepsilon > 0$. One can extract ℓ bits of ε-secure randomness from $\rho_A^{\otimes n}$ whenever

$$\frac{\ell}{n} \leqslant 2\log(d_A) - \min_{\alpha\in(1,2]} \left(H_{\frac{\alpha}{2\alpha-1}}(A) - \frac{\alpha}{n(\alpha-1)} \log(\varepsilon) \right) + \frac{2}{n}.$$

- Quantifies how useful a given source is for QRNG

Main result

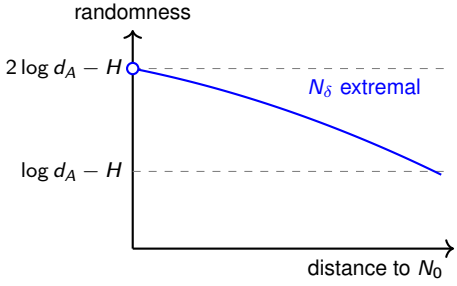
Theorem

Let ρ_A be a d_A-dimensional quantum state and $\varepsilon > 0$. One can extract ℓ bits of ε-secure randomness from $\rho_A^{\otimes n}$ whenever

$$\frac{\ell}{n} \leqslant 2\log(d_A) - \min_{\alpha\in(1,2]} \left(H_{\frac{\alpha}{2\alpha-1}}(A) - \frac{\alpha}{n(\alpha-1)} \log(\varepsilon) \right) + \frac{2}{n}.$$

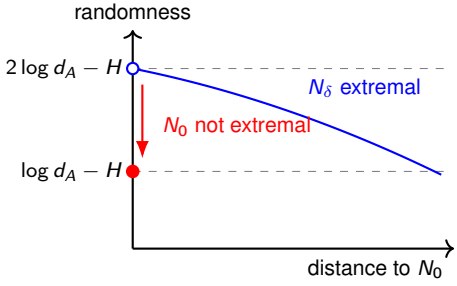
- Quantifies how useful a given source is for QRNG
- Converges to the asymptotic rate $2\log(d_A) - H(A)$

A caveat: discontinuities



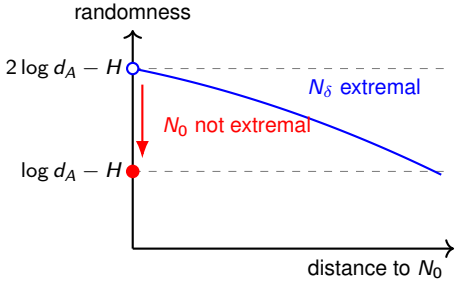
- Optimum approached by extremal rank-1 POVMs $N_\delta \to N_0 = \left\{ \frac{1}{d_A} |y\rangle\langle y| \right\}_{x,y}$

A caveat: discontinuities



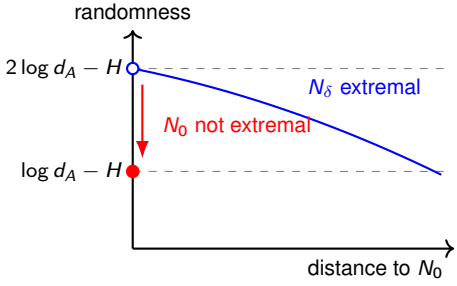
- Optimum approached by extremal rank-1 POVMs $N_\delta \to N_0 = \left\{ \frac{1}{d_A} |y\rangle\langle y| \right\}_{x,y}$
- N_0 is not extremal: randomness drops by $\log(d_A)$ bits!

A caveat: discontinuities



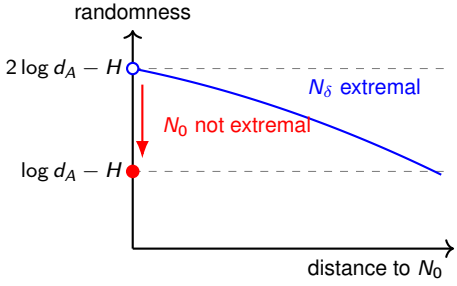
- Optimum approached by extremal rank-1 POVMs $N_\delta \to N_0 = \left\{ \frac{1}{d_A} |y\rangle\langle y| \right\}_{x,y}$
- N_0 is not extremal: randomness drops by $\log(d_A)$ bits!
- Attack: the device implements N_0 instead of N_δ, undetectable as $\delta \to 0$

A caveat: discontinuities



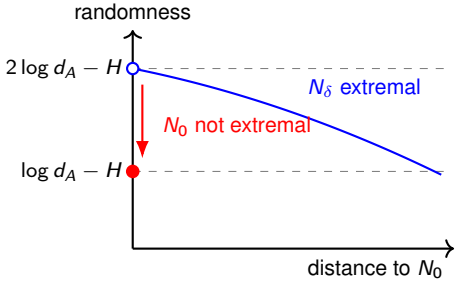
- Optimum approached by extremal rank-1 POVMs $N_\delta \to N_0 = \left\{ \frac{1}{d_A} |y\rangle\langle y| \right\}_{x,y}$
- N_0 is not extremal: randomness drops by $\log(d_A)$ bits!
- Attack: the device implements N_0 instead of N_δ, undetectable as $\delta \to 0$
- Continuous on extremal rank-1 POVMs, and on full-rank POVMs

A caveat: discontinuities



- Optimum approached by extremal rank-1 POVMs $N_\delta \to N_0 = \left\{ \frac{1}{d_A} |y\rangle\langle y| \right\}_{x,y}$
- N_0 is not extremal: randomness drops by $\log(d_A)$ bits!
- Attack: the device implements N_0 instead of N_δ, undetectable as $\delta \to 0$
- Continuous on extremal rank-1 POVMs, and on full-rank POVMs
- Known feasible optimal point in all dimensions

A caveat: discontinuities



- Optimum approached by extremal rank-1 POVMs $N_\delta \to N_0 = \left\{ \frac{1}{d_A} |y\rangle\langle y| \right\}_{x,y}$
- N_0 is not extremal: randomness drops by $\log(d_A)$ bits!
- Attack: the device implements N_0 instead of N_δ, undetectable as $\delta \to 0$
- Continuous on extremal rank-1 POVMs, and on full-rank POVMs
- Known feasible optimal point in all dimensions
- Open: where are all the discontinuities? Is the optimal POVM far from them?

Outline of this talk

Chapter 3

Quantum random number generation

How much secure randomness is in a quantum state?

(Nemoz, Anco and Brown, 2024)

Chapter 4

Real-valued pseudorandom states

Do complex amplitudes increase the security of pseudorandom states?

(Nemoz, Alléaume and Brown, 2026)

Chapter 5

Pseudorandom coherent states

Can pseudorandom states travel on light?

(Kuzaka, Nemoz and Alléaume, in preparation)

Pseudorandom quantum states (PRS)

Definition (Ji et al., 2018), informal

A keyed family $\{|\psi_k\rangle\}_{k\in\{0,1\}^\lambda}$ of d-dimensional states that is

Pseudorandom quantum states (PRS)

Definition (Ji et al., 2018), informal

A keyed family $\{|\psi_k\rangle\}_{k\in\{0,1\}^\lambda}$ of d-dimensional states that is

- efficiently preparable from k,

Pseudorandom quantum states (PRS)

Definition (Ji et al., 2018), informal

A keyed family $\{|\psi_k\rangle\}_{k\in\{0,1\}^\lambda}$ of d-dimensional states that is

- efficiently preparable from k,
- indistinguishable from Haar-random states by any efficient adversary, even given $t = \mathrm{poly}(\lambda)$ copies.

Pseudorandom quantum states (PRS)

Definition (Ji et al., 2018), informal

A keyed family $\{|\psi_k\rangle\}_{k\in\{0,1\}^\lambda}$ of d-dimensional states that is

- efficiently preparable from k,
- indistinguishable from Haar-random states by any efficient adversary, even given $t = \mathrm{poly}(\lambda)$ copies.

Used to perform cryptography in Microcrypt.

Pseudorandom quantum states (PRS)

Definition (Ji et al., 2018), informal

A keyed family $\{|\psi_k\rangle\}_{k\in\{0,1\}^\lambda}$ of d-dimensional states that is

- efficiently preparable from k,
- indistinguishable from Haar-random states by any efficient adversary, even given $t = \mathrm{poly}(\lambda)$ copies.

Used to perform cryptography in Microcrypt. Typical security proof, e.g. for binary phase states (Ji et al., 2018; Brakerski and Shmueli, 2019) with $\{f_k\}_k$ a quantum-secure PRF:

Pseudorandom quantum states (PRS)

Definition (Ji et al., 2018), informal

A keyed family $\{|\psi_k\rangle\}_{k\in\{0,1\}^\lambda}$ of d-dimensional states that is

- efficiently preparable from k,
- indistinguishable from Haar-random states by any efficient adversary, even given $t = \mathrm{poly}(\lambda)$ copies.

Used to perform cryptography in Microcrypt. Typical security proof, e.g. for binary phase states (Ji et al., 2018; Brakerski and Shmueli, 2019) with $\{f_k\}_k$ a quantum-secure PRF:

$$\underbrace{\left(\frac{1}{\sqrt{d}}\sum_x (-1)^{f_k(x)}|x\rangle\right)^{\otimes t}}_{\text{PRS}} \approx_c^{\text{qPRF}} \underbrace{\left(\frac{1}{\sqrt{d}}\sum_x (-1)^{f(x)}|x\rangle\right)^{\otimes t}}_{f \text{ truly random}} \approx_{\varepsilon(t,d)}^{s} t \text{ copies of a Haar random state}$$

Advantage $\leqslant \mathrm{negl}(\lambda) + \varepsilon(t,d)$: ε limits the number of copies

Real versus complex amplitudes

ε-approximate state t-design

An ensemble $(\mathcal{S}, \mu)$ such that $\frac{1}{2}\left\|\mathbb{E}_{|\psi\rangle\sim\mu}\left[|\psi\rangle\langle\psi|^{\otimes t}\right] - \rho_{\mathcal{U}_d^t}\right\|_1 \leqslant \varepsilon$, where $\rho_{\mathcal{U}_d^t} \stackrel{\text{def}}{=} \mathbb{E}_{|\psi\rangle\sim\text{Haar}}\left[|\psi\rangle\langle\psi|^{\otimes t}\right]$.

Real versus complex amplitudes

ε-approximate state t-design

An ensemble $(\mathcal{S}, \mu)$ such that $\frac{1}{2}\left\|\mathbb{E}_{|\psi\rangle\sim\mu}\left[|\psi\rangle\langle\psi|^{\otimes t}\right] - \rho_{\mathcal{U}_d^t}\right\|_1 \leqslant \varepsilon$, where $\rho_{\mathcal{U}_d^t} \stackrel{\text{def}}{=} \mathbb{E}_{|\psi\rangle\sim\text{Haar}}\left[|\psi\rangle\langle\psi|^{\otimes t}\right]$.

Real versus complex amplitudes

ε-approximate state t-design

An ensemble $(\mathcal{S}, \mu)$ such that $\frac{1}{2}\left\|\mathbb{E}_{|\psi\rangle\sim\mu}\left[|\psi\rangle\langle\psi|^{\otimes t}\right] - \rho_{\mathcal{U}_d^t}\right\|_1 \leqslant \varepsilon$, where $\rho_{\mathcal{U}_d^t} \stackrel{\text{def}}{=} \mathbb{E}_{|\psi\rangle\sim\text{Haar}}\left[|\psi\rangle\langle\psi|^{\otimes t}\right]$.

	Real amplitudes	Complex amplitudes
Best known ε	$\Theta\left(\frac{t^2}{d}\right)$ (Brakerski and Shmueli, 2019; Jeronimo et al., 2024; Cui et al., 2025)	$\Theta\left(\frac{t}{d}\right)$ (Brakerski and Shmueli, 2020; Lu et al., 2

Real versus complex amplitudes

ε-approximate state t-design

An ensemble $(\mathcal{S}, \mu)$ such that $\frac{1}{2}\left\|\mathbb{E}_{|\psi\rangle\sim\mu}\left[|\psi\rangle\langle\psi|^{\otimes t}\right] - \rho_{\mathcal{U}_d^t}\right\|_1 \leqslant \varepsilon$, where $\rho_{\mathcal{U}_d^t} \stackrel{\text{def}}{=} \mathbb{E}_{|\psi\rangle\sim\text{Haar}}\left[|\psi\rangle\langle\psi|^{\otimes t}\right]$.

	Real amplitudes	Complex amplitudes
Best known ε	$\Theta\left(\frac{t^2}{d}\right)$ (Brakerski and Shmueli, 2019; Jeronimo et al., 2024; Cui et al., 2025)	$\Theta\left(\frac{t}{d}\right)$ (Brakerski and Shmueli, 2020; Lu et al., 2

Is $\frac{t^2}{d}$ a fundamental limit for real-valued designs?

Real versus complex amplitudes

ε-approximate state t-design

An ensemble $(\mathcal{S}, \mu)$ such that $\frac{1}{2}\left\|\mathbb{E}_{|\psi\rangle\sim\mu}\left[|\psi\rangle\langle\psi|^{\otimes t}\right] - \rho_{\mathcal{U}_d^t}\right\|_1 \leqslant \varepsilon$, where $\rho_{\mathcal{U}_d^t} \stackrel{\text{def}}{=} \mathbb{E}_{|\psi\rangle\sim\text{Haar}}\left[|\psi\rangle\langle\psi|^{\otimes t}\right]$.

	Real amplitudes	Complex amplitudes
Best known ε	$\Theta\left(\frac{t^2}{d}\right)$ (Brakerski and Shmueli, 2019; Jeronimo et al., 2024; Cui et al., 2025)	$\Theta\left(\frac{t}{d}\right)$ (Brakerski and Shmueli, 2020; Lu et al., 2

Is $\frac{t^2}{d}$ a fundamental limit for real-valued designs? **Yes.**

Step 1: the best design with fixed imaginarity

Proposition

Let

$$\rho_{\mathcal{O}_d^t} \stackrel{\text{def}}{=} \int_{O \in \mathcal{O}_d} \left(O \left|0\right\rangle\!\left\langle 0\right| O^\top \right)^{\otimes t} \mathrm{d}\mu(O) .$$

Step 1: the best design with fixed imaginarity

Proposition

Let

$$\rho_{\mathcal{O}_d^t} \stackrel{\text{def}}{=} \int_{O \in \mathcal{O}_d} \left(O \left|0\right\rangle\!\left\langle 0\right| O^\top \right)^{\otimes t} \mathrm{d}\mu(O)\,.$$

For any ensemble $\mathcal{S}$ of real-valued states, with

$$\rho_{\mathcal{S}^t} \stackrel{\text{def}}{=} \mathbb{E}_{|\psi\rangle \sim \mathcal{S}} \left[|\psi\rangle\!\langle\psi|^{\otimes t} \right]$$

Step 1: the best design with fixed imaginarity

Proposition

Let

$$\rho_{\mathcal{O}_d^t} \stackrel{\text{def}}{=} \int_{O \in \mathcal{O}_d} \left(O \left|0\right\rangle\!\left\langle 0\right| O^\top \right)^{\otimes t} \mathrm{d}\mu(O)\,.$$

For any ensemble $\mathcal{S}$ of real-valued states, with

$$\rho_{\mathcal{S}^t} \stackrel{\text{def}}{=} \mathbb{E}_{|\psi\rangle \sim \mathcal{S}} \left[|\psi\rangle\!\langle\psi|^{\otimes t} \right]$$

we have

$$\left\| \rho_{\mathcal{S}^t} - \rho_{\mathcal{U}_d^t} \right\| \geqslant \left\| \rho_{\mathcal{O}_d^t} - \rho_{\mathcal{U}_d^t} \right\| .$$

Step 1: the best design with fixed imaginarity

Proposition

Let

$$\rho_{\mathcal{O}_d^t} \stackrel{\text{def}}{=} \int_{O \in \mathcal{O}_d} \left(O \left|0\right\rangle\!\left\langle 0\right| O^\top \right)^{\otimes t} \mathrm{d}\mu(O)\,.$$

For any ensemble $\mathcal{S}$ of real-valued states, with

$$\rho_{\mathcal{S}^t} \stackrel{\text{def}}{=} \mathbb{E}_{|\psi\rangle \sim \mathcal{S}} \left[|\psi\rangle\!\langle\psi|^{\otimes t} \right]$$

we have

$$\left\| \rho_{\mathcal{S}^t} - \rho_{\mathcal{U}_d^t} \right\| \geqslant \left\| \rho_{\mathcal{O}_d^t} - \rho_{\mathcal{U}_d^t} \right\| .$$

$\implies \frac{1}{2} \left\| \rho_{\mathcal{O}_d^t} - \rho_{\mathcal{U}_d^t} \right\|$ is the best possible ε for real-valued states

Step 2: diagonalizing with symmetries

Schur's lemma, informally

If M has symmetries ($VMV^\dagger = M$ for a group of unitaries V), its eigenvalues come in **blocks**: M is constant on each block.
The **more** symmetries, the **larger** the blocks.

Step 2: diagonalizing with symmetries

Schur's lemma, informally

If M has symmetries ($VMV^\dagger = M$ for a group of unitaries V), its eigenvalues come in **blocks**: M is constant on each block.
The **more** symmetries, the **larger** the blocks.

Both states live on the symmetric subspace $\vee^t \mathbb{C}^d$, spanned by the $|\psi\rangle^{\otimes t}$

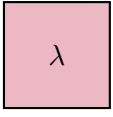
Step 2: diagonalizing with symmetries

Schur's lemma, informally

If M has symmetries ($VMV^\dagger = M$ for a group of unitaries V), its eigenvalues come in **blocks**: M is constant on each block.
The **more** symmetries, the **larger** the blocks.

Both states live on the symmetric subspace $\vee^t \mathbb{C}^d$, spanned by the $|\psi\rangle^{\otimes t}$

Complex: $\rho_{\mathcal{U}_d^t}$ commutes with every $U^{\otimes t}$: **enough** symmetries for a single block



$$\rho_{\mathcal{U}_d^t} = \binom{d+t-1}{t}^{-1} \Pi_{\vee^t \mathbb{C}^d}$$

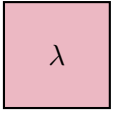
Step 2: diagonalizing with symmetries

Schur's lemma, informally

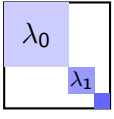
If M has symmetries ($VMV^\dagger = M$ for a group of unitaries V), its eigenvalues come in **blocks**: M is constant on each block.
The **more** symmetries, the **larger** the blocks.

Both states live on the symmetric subspace $\vee^t \mathbb{C}^d$, spanned by the $|\psi\rangle^{\otimes t}$

Complex: $\rho_{\mathcal{U}_d^t}$ commutes with every $U^{\otimes t}$: **enough** symmetries for a single block



Real: $\rho_{\mathcal{O}_d^t}$ only commutes with the $O^{\otimes t}$: a **subset** of them, so smaller blocks



$$\rho_{\mathcal{U}_d^t} = \binom{d+t-1}{t}^{-1} \Pi_{\vee^t \mathbb{C}^d}$$

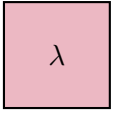
Step 2: diagonalizing with symmetries

Schur's lemma, informally

If M has symmetries ($VMV^\dagger = M$ for a group of unitaries V), its eigenvalues come in **blocks**: M is constant on each block.
The **more** symmetries, the **larger** the blocks.

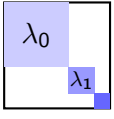
Both states live on the symmetric subspace $\vee^t \mathbb{C}^d$, spanned by the $|\psi\rangle^{\otimes t}$

Complex: $\rho_{\mathcal{U}_d^t}$ commutes with every $U^{\otimes t}$: **enough** symmetries for a single block



$$\rho_{\mathcal{U}_d^t} = \binom{d+t-1}{t}^{-1} \Pi_{\vee^t \mathbb{C}^d}$$

Real: $\rho_{\mathcal{O}_d^t}$ only commutes with the $O^{\otimes t}$: a **subset** of them, so smaller blocks



Goal: identify the blocks (and the λ_k)

Finding the blocks

We're looking for the subspaces that are invariant by applying $O^{\otimes t}$ for all $O \in \mathcal{O}_d$.

Finding the blocks

We're looking for the subspaces that are invariant by applying $O^{\otimes t}$ for all $O \in \mathcal{O}_d$. Take $t = 2$ and $|\Phi\rangle \stackrel{\text{def}}{=} \frac{1}{\sqrt{d}} \sum_x |x, x\rangle \in \vee^2 \mathbb{C}^d$.

Finding the blocks

We're looking for the subspaces that are invariant by applying $O^{\otimes t}$ for all $O \in \mathcal{O}_d$. Take $t = 2$ and $|\Phi\rangle \stackrel{\text{def}}{=} \frac{1}{\sqrt{d}} \sum_x |x, x\rangle \in \vee^2 \mathbb{C}^d$. Then $O^{\otimes 2} |\Phi\rangle = |\Phi\rangle$ for every $O \in \mathcal{O}_d$, but $U^{\otimes 2} |\Phi\rangle \neq |\Phi\rangle$ doesn't hold for every $U \in \mathcal{U}_d$.

Finding the blocks

We're looking for the subspaces that are invariant by applying $O^{\otimes t}$ for all $O \in \mathcal{O}_d$. Take $t = 2$ and $|\Phi\rangle \stackrel{\text{def}}{=} \frac{1}{\sqrt{d}} \sum_x |x, x\rangle \in \vee^2 \mathbb{C}^d$. Then $O^{\otimes 2} |\Phi\rangle = |\Phi\rangle$ for every $O \in \mathcal{O}_d$, but $U^{\otimes 2} |\Phi\rangle \neq |\Phi\rangle$ doesn't hold for every $U \in \mathcal{U}_d$.

- $|\Phi\rangle\langle\Phi|$ is a block on its own for $\rho_{\mathcal{O}_d^2}$, but just part of the single block of $\rho_{\mathcal{U}_d^2}$

Finding the blocks

We're looking for the subspaces that are invariant by applying $O^{\otimes t}$ for all $O \in \mathcal{O}_d$. Take $t = 2$ and $|\Phi\rangle \overset{\text{def}}{=} \frac{1}{\sqrt{d}} \sum_x |x, x\rangle \in \vee^2 \mathbb{C}^d$. Then $O^{\otimes 2} |\Phi\rangle = |\Phi\rangle$ for every $O \in \mathcal{O}_d$, but $U^{\otimes 2} |\Phi\rangle \neq |\Phi\rangle$ doesn't hold for every $U \in \mathcal{U}_d$.

- $|\Phi\rangle\langle\Phi|$ is a block on its own for $\rho_{\mathcal{O}_d^2}$, but just part of the single block of $\rho_{\mathcal{U}_d^2}$
- For larger t: many blocks, of many sizes, in dimension $\binom{d+t-1}{t}$

Finding the blocks

We're looking for the subspaces that are invariant by applying $O^{\otimes t}$ for all $O \in \mathcal{O}_d$. Take $t = 2$ and $|\Phi\rangle \stackrel{\text{def}}{=} \frac{1}{\sqrt{d}} \sum_x |x, x\rangle \in \vee^2 \mathbb{C}^d$. Then $O^{\otimes 2} |\Phi\rangle = |\Phi\rangle$ for every $O \in \mathcal{O}_d$, but $U^{\otimes 2} |\Phi\rangle \neq |\Phi\rangle$ doesn't hold for every $U \in \mathcal{U}_d$.

- $|\Phi\rangle\langle\Phi|$ is a block on its own for $\rho_{\mathcal{O}_d^2}$, but just part of the single block of $\rho_{\mathcal{U}_d^2}$
- For larger t: many blocks, of many sizes, in dimension $\binom{d+t-1}{t}$
- How do we find them?

Our approach: from tensors to polynomials

- Isomorphism $\vee^t \mathbb{C}^d \cong \mathcal{P}_d^t$, homogeneous polynomials of degree t with d variables.

Our approach: from tensors to polynomials

- Isomorphism $\vee^t \mathbb{C}^d \cong \mathcal{P}_d^t$, homogeneous polynomials of degree t with d variables. If $P(X)$ is the image of $|\psi\rangle$, then

Our approach: from tensors to polynomials

- Isomorphism $\vee^t \mathbb{C}^d \cong \mathcal{P}_d^t$, homogeneous polynomials of degree t with d variables. If $P(X)$ is the image of $|\psi\rangle$, then
  - the Bombieri norm (Bea+90) of $P(X)$ is the usual norm of $|\psi\rangle$

Our approach: from tensors to polynomials

- Isomorphism $\vee^t \mathbb{C}^d \cong \mathcal{P}_d^t$, homogeneous polynomials of degree t with d variables. If $P(X)$ is the image of $|\psi\rangle$, then
  - the Bombieri norm (**Bea+90**) of $P(X)$ is the usual norm of $|\psi\rangle$
  - the image of $O^{\otimes t}|\psi\rangle$ is $P\left(O^\top X\right)$

Our approach: from tensors to polynomials

- Isomorphism $\vee^t \mathbb{C}^d \cong \mathcal{P}_d^t$, homogeneous polynomials of degree t with d variables. If $P(X)$ is the image of $|\psi\rangle$, then
  - the Bombieri norm (**Bea+90**) of $P(X)$ is the usual norm of $|\psi\rangle$
  - the image of $O^{\otimes t}|\psi\rangle$ is $P\left(O^\top X\right)$
- Blocks: harmonic decomposition (Coifman and Weiss, 1968), with *harmonic* polynomials $\mathcal{H}_d^n \stackrel{\text{def}}{=} \{H : \Delta H = 0\}$

$$\mathcal{P}_d^t = \mathcal{H}_d^t \oplus \|X\|^2 \mathcal{H}_d^{t-2} \oplus \|X\|^4 \mathcal{H}_d^{t-4} \oplus \cdots$$

with

$$\|X\|^2 = \left(\sum_{i=1}^d X_i\right)^2 .$$

Our approach: from tensors to polynomials

- Isomorphism $\vee^t \mathbb{C}^d \cong \mathcal{P}_d^t$, homogeneous polynomials of degree t with d variables. If $P(X)$ is the image of $|\psi\rangle$, then
  - the Bombieri norm (Bea+90) of $P(X)$ is the usual norm of $|\psi\rangle$
  - the image of $O^{\otimes t}|\psi\rangle$ is $P\left(O^\top X\right)$
- Blocks: harmonic decomposition (Coifman and Weiss, 1968), with *harmonic* polynomials $\mathcal{H}_d^n \stackrel{\text{def}}{=} \{H : \Delta H = 0\}$

$$\mathcal{P}_d^t = \mathcal{H}_d^t \oplus \|X\|^2 \mathcal{H}_d^{t-2} \oplus \|X\|^4 \mathcal{H}_d^{t-4} \oplus \cdots$$

with

$$\|X\|^2 = \left(\sum_{i=1}^d X_i\right)^2 .$$

- Eigenvalues:

$$\mathrm{Tr}\left[\rho_{\mathcal{O}_d^t} \Pi_k\right] = \lambda_{d,t,k} \,\mathrm{Tr}\left[\Pi_k\right]$$

Our approach: from tensors to polynomials

- Isomorphism $\vee^t \mathbb{C}^d \cong \mathcal{P}_d^t$, homogeneous polynomials of degree t with d variables. If $P(X)$ is the image of $|\psi\rangle$, then
  - the Bombieri norm (Bea+90) of $P(X)$ is the usual norm of $|\psi\rangle$
  - the image of $O^{\otimes t}|\psi\rangle$ is $P\left(O^\top X\right)$
- Blocks: harmonic decomposition (Coifman and Weiss, 1968), with *harmonic* polynomials $\mathcal{H}_d^n \stackrel{\text{def}}{=} \{H : \Delta H = 0\}$

$$\mathcal{P}_d^t = \mathcal{H}_d^t \oplus \|X\|^2 \mathcal{H}_d^{t-2} \oplus \|X\|^4 \mathcal{H}_d^{t-4} \oplus \cdots$$

with

$$\|X\|^2 = \left(\sum_{i=1}^d X_i\right)^2 .$$

- Eigenvalues:

$$\mathrm{Tr}\left[\rho_{\mathcal{O}_d^t} \Pi_k\right] = \lambda_{d,t,k} \underbrace{\mathrm{Tr}\left[\Pi_k\right]}_{\dim\left(\mathcal{H}_d^{t-2k}\right)}$$

Our approach: from tensors to polynomials

- Isomorphism $\vee^t \mathbb{C}^d \cong \mathcal{P}_d^t$, homogeneous polynomials of degree t with d variables. If $P(X)$ is the image of $|\psi\rangle$, then
  - the Bombieri norm (**Bea+90**) of $P(X)$ is the usual norm of $|\psi\rangle$
  - the image of $O^{\otimes t}|\psi\rangle$ is $P\left(O^\top X\right)$
- Blocks: harmonic decomposition (Coifman and Weiss, 1968), with *harmonic* polynomials $\mathcal{H}_d^n \stackrel{\text{def}}{=} \{H : \Delta H = 0\}$

$$\mathcal{P}_d^t = \mathcal{H}_d^t \oplus \|X\|^2 \mathcal{H}_d^{t-2} \oplus \|X\|^4 \mathcal{H}_d^{t-4} \oplus \cdots$$

with

$$\|X\|^2 = \left(\sum_{i=1}^d X_i\right)^2.$$

- Eigenvalues:

$$\mathrm{Tr}\left[\rho_{\mathcal{O}_d^t} \Pi_k\right] = \lambda_{d,t,k} \dim\left(\mathcal{H}_d^{t-2k}\right)$$

Our approach: from tensors to polynomials

- Isomorphism $\vee^t \mathbb{C}^d \cong \mathcal{P}_d^t$, homogeneous polynomials of degree t with d variables. If $P(X)$ is the image of $|\psi\rangle$, then
  - the Bombieri norm (**Bea+90**) of $P(X)$ is the usual norm of $|\psi\rangle$
  - the image of $O^{\otimes t} |\psi\rangle$ is $P\left(O^\top X\right)$
- Blocks: harmonic decomposition (Coifman and Weiss, 1968), with *harmonic* polynomials $\mathcal{H}_d^n \stackrel{\text{def}}{=} \{H : \Delta H = 0\}$

$$\mathcal{P}_d^t = \mathcal{H}_d^t \oplus \|X\|^2 \mathcal{H}_d^{t-2} \oplus \|X\|^4 \mathcal{H}_d^{t-4} \oplus \cdots$$

with

$$\|X\|^2 = \left(\sum_{i=1}^d X_i\right)^2 .$$

- Eigenvalues:

$$\underbrace{\mathrm{Tr}\left[\rho_{\mathcal{O}_d^t}\right]}_{\|\Pi_k P_\delta\|^2_{\text{Bombieri}}} = \lambda_{d,t,k} \dim\left(\mathcal{H}_d^{t-2k}\right)$$

Our approach: from tensors to polynomials

- Isomorphism $\vee^t \mathbb{C}^d \cong \mathcal{P}_d^t$, homogeneous polynomials of degree t with d variables. If $P(X)$ is the image of $|\psi\rangle$, then
  - the Bombieri norm (**Bea+90**) of $P(X)$ is the usual norm of $|\psi\rangle$
  - the image of $O^{\otimes t} |\psi\rangle$ is $P\left(O^\top X\right)$
- Blocks: harmonic decomposition (Coifman and Weiss, 1968), with *harmonic* polynomials $\mathcal{H}_d^n \stackrel{\text{def}}{=} \{H \,:\, \Delta H = 0\}$

$$\mathcal{P}_d^t = \mathcal{H}_d^t \oplus \|X\|^2 \mathcal{H}_d^{t-2} \oplus \|X\|^4 \mathcal{H}_d^{t-4} \oplus \cdots$$

with

$$\|X\|^2 = \left(\sum_{i=1}^{d} X_i\right)^2 .$$

- Eigenvalues:

$$\|\Pi_k P_\delta\|^2_{\text{Bombieri}} = \lambda_{d,t,k} \dim\left(\mathcal{H}_d^{t-2k}\right)$$

Our approach: from tensors to polynomials

- Isomorphism $\vee^t \mathbb{C}^d \cong \mathcal{P}_d^t$, homogeneous polynomials of degree t with d variables. If $P(X)$ is the image of $|\psi\rangle$, then
  - the Bombieri norm (**Bea+90**) of $P(X)$ is the usual norm of $|\psi\rangle$
  - the image of $O^{\otimes t}|\psi\rangle$ is $P\left(O^\top X\right)$
- Blocks: harmonic decomposition (Coifman and Weiss, 1968), with *harmonic* polynomials $\mathcal{H}_d^n \stackrel{\text{def}}{=} \{H : \Delta H = 0\}$

$$\mathcal{P}_d^t = \mathcal{H}_d^t \oplus \|X\|^2 \mathcal{H}_d^{t-2} \oplus \|X\|^4 \mathcal{H}_d^{t-4} \oplus \cdots$$

with

$$\|X\|^2 = \left(\sum_{i=1}^d X_i\right)^2.$$

- Eigenvalues:

$$\|\Pi_k P_\delta\|^2_{\text{Bombieri}} = \lambda_{d,t,k} \dim\left(\mathcal{H}_d^{t-2k}\right)$$

$\Longrightarrow$ We can get a closed-form expression for the eigenvalues!

Main result: real amplitudes cost a factor t

Theorem

For $t \lesssim \sqrt{2d \ln d}$:

$$\frac{1}{2}\left\|\rho_{\mathcal{O}_d^t 0} - \rho_{\mathcal{U}_d^t}\right\|_1 = \frac{t(t-1)}{2d} + \Theta\left(\frac{t^3}{d^2}\right).$$

Main result: real amplitudes cost a factor t

Theorem

For $t \lesssim \sqrt{2d \ln d}$:

$$\frac{1}{2}\left\|\rho_{\mathcal{O}_d^t 0} - \rho_{\mathcal{U}_d^t}\right\|_1 = \frac{t(t-1)}{2d} + \Theta\left(\frac{t^3}{d^2}\right).$$

$\Longrightarrow$ **Complex amplitudes are necessary** to beat $\varepsilon = \Theta\left(\frac{t^2}{d}\right)$

From designs to pseudorandom states?

- Optimal measurement $\{P, I - P\}$ distinguishing $\rho_{\mathcal{O}_d^t}$ from $\rho_{\mathcal{U}_d^t}$: P projects onto the blocks where $\lambda_{d,t,k} \leqslant \binom{d+t-1}{t}^{-1}$

From designs to pseudorandom states?

- Optimal measurement $\{P, I - P\}$ distinguishing $\rho_{\mathcal{O}_d^t}$ from $\rho_{\mathcal{U}_d^t}$: P projects onto the blocks where $\lambda_{d,t,k} \leqslant \binom{d+t-1}{t}^{-1}$
- Works for *any* real ensemble $\mathcal{S}$: $\mathrm{Tr}\left[P\rho_{\mathcal{S}}^{(t)}\right] = \mathrm{Tr}\left[P\rho_{\mathcal{O}_d^t}\right]$

From designs to pseudorandom states?

- Optimal measurement $\{P, I - P\}$ distinguishing $\rho_{\mathcal{O}_d^t}$ from $\rho_{\mathcal{U}_d^t}$: P projects onto the blocks where $\lambda_{d,t,k} \leqslant \binom{d+t-1}{t}^{-1}$
- Works for *any* real ensemble $\mathcal{S}$: $\mathrm{Tr}\left[P\rho_{\mathcal{S}}^{(t)}\right] = \mathrm{Tr}\left[P\rho_{\mathcal{O}_d^t}\right]$
- This bounds ε, not the PRS advantage: is P *efficient*?

From designs to pseudorandom states?

- Optimal measurement $\{P, I - P\}$ distinguishing $\rho_{\mathcal{O}_d^t}$ from $\rho_{\mathcal{U}_d^t}$: P projects onto the blocks where $\lambda_{d,t,k} \leqslant \binom{d+t-1}{t}^{-1}$
- Works for *any* real ensemble $\mathcal{S}$: $\mathrm{Tr}\left[P\rho_{\mathcal{S}}^{(t)}\right] = \mathrm{Tr}\left[P\rho_{\mathcal{O}_d^t}\right]$
- This bounds ε, not the PRS advantage: is P *efficient*?

Open problem

Can $\{P, I - P\}$ be implemented to precision η in time $\mathrm{poly}\left(t, \log d, \frac{1}{\eta}\right)$?

From designs to pseudorandom states?

- Optimal measurement $\{P, I - P\}$ distinguishing $\rho_{\mathcal{O}_d^t}$ from $\rho_{\mathcal{U}_d^t}$: P projects onto the blocks where $\lambda_{d,t,k} \leqslant \binom{d+t-1}{t}^{-1}$
- Works for *any* real ensemble $\mathcal{S}$: $\mathrm{Tr}\left[P\rho_{\mathcal{S}}^{(t)}\right] = \mathrm{Tr}\left[P\rho_{\mathcal{O}_d^t}\right]$
- This bounds ε, not the PRS advantage: is P *efficient*?

Open problem

Can $\{P, I - P\}$ be implemented to precision η in time $\mathrm{poly}\left(t, \log d, \frac{1}{\eta}\right)$?

- Route: Schur transform (Burchardt et al., 2025), then change of basis from monomials to harmonics

From designs to pseudorandom states?

- Optimal measurement $\{P, I - P\}$ distinguishing $\rho_{\mathcal{O}_d^t}$ from $\rho_{\mathcal{U}_d^t}$: P projects onto the blocks where $\lambda_{d,t,k} \leqslant \binom{d+t-1}{t}^{-1}$
- Works for *any* real ensemble $\mathcal{S}$: $\mathrm{Tr}\left[P\rho_{\mathcal{S}}^{(t)}\right] = \mathrm{Tr}\left[P\rho_{\mathcal{O}_d^t}\right]$
- This bounds ε, not the PRS advantage: is P *efficient*?

Open problem

Can $\{P, I - P\}$ be implemented to precision η in time $\mathrm{poly}\left(t, \log d, \frac{1}{\eta}\right)$?

- Route: Schur transform (Burchardt et al., 2025), then change of basis from monomials to harmonics
- If so (we conjecture it): **real-valued PRS** are insecure beyond $\sim \sqrt{d}$ copies

Byproducts of the spectral decomposition

- Exact designs from orthogonal twirling a state with imaginarity following a Beta law

Byproducts of the spectral decomposition

- Exact designs from orthogonal twirling a state with imaginarity following a Beta law
- Imaginarity testing requires $\Omega\left(\sqrt{d}\right)$ copies

Byproducts of the spectral decomposition

- Exact designs from orthogonal twirling a state with imaginarity following a Beta law
- Imaginarity testing requires $\Omega\left(\sqrt{d}\right)$ copies
- Optimal measurement for imaginarity estimation, saving copies over (Haug et al., 2025)

Byproducts of the spectral decomposition

- Exact designs from orthogonal twirling a state with imaginarity following a Beta law
- Imaginarity testing requires $\Omega\left(\sqrt{d}\right)$ copies
- Optimal measurement for imaginarity estimation, saving copies over (Haug et al., 2025)
- Real-valued state estimation: the Pretty Good Measurement is optimal, with infidelity $\sim \frac{d-1}{4t}$ (numerically) vs. $\frac{d-1}{t}$ for complex states

Outline of this talk

Chapter 3

Quantum random number generation

How much secure randomness is in a quantum state?

(Nemoz, Anco and Brown, 2024)

Chapter 4

Real-valued pseudorandom states

Do complex amplitudes increase the security of pseudorandom states?

(Nemoz, Alléaume and Brown, 2026)

Chapter 5

Pseudorandom coherent states

Can pseudorandom states travel on light?

(Kuzaka, Nemoz and Alléaume, in preparation)

Pseudorandom states meet optics

- PRS live in the qudit model: $|\psi\rangle \in \mathbb{C}^d$, given t times...

Pseudorandom states meet optics

- PRS live in the qudit model: $|\psi\rangle \in \mathbb{C}^d$, given t times...
- ...but quantum communication uses light: coherent states

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{N \geqslant 0} \frac{\alpha^N}{\sqrt{N!}} |N\rangle\,, \qquad \text{number of photons } N \sim \text{Poisson}\left(|\alpha|^2\right)$$

Pseudorandom states meet optics

- PRS live in the qudit model: $|\psi\rangle \in \mathbb{C}^d$, given t times...
- ...but quantum communication uses light: coherent states

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{N \geqslant 0} \frac{\alpha^N}{\sqrt{N!}} |N\rangle\,, \qquad \text{number of photons } N \sim \text{Poisson}\left(|\alpha|^2\right)$$

- No Haar measure on the set of displacement operators!

Pseudorandom states meet optics

- PRS live in the qudit model: $|\psi\rangle \in \mathbb{C}^d$, given t times...
- ...but quantum communication uses light: coherent states

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{N \geqslant 0} \frac{\alpha^N}{\sqrt{N!}} |N\rangle\,, \qquad \text{number of photons } N \sim \text{Poisson}\left(|\alpha|^2\right)$$

- No Haar measure on the set of displacement operators!

  What is a *uniformly random* coherent state?

Pseudorandom states meet optics

- PRS live in the qudit model: $|\psi\rangle \in \mathbb{C}^d$, given t times...
- ...but quantum communication uses light: coherent states

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{N \geqslant 0} \frac{\alpha^N}{\sqrt{N!}} |N\rangle\,, \qquad \text{number of photons } N \sim \text{Poisson}\left(|\alpha|^2\right)$$

- No Haar measure on the set of displacement operators!

What is a *uniformly random* coherent state?

Mapping of Arrazola and Lütkenhaus (2014): one mode per basis state

$$\sum_{x=0}^{d-1} \psi_x |x\rangle \longmapsto \bigotimes_{x=0}^{d-1} |\alpha\, \psi_x\rangle$$

Haar-random coherent states

Average the mapped state over Haar-random $|\psi\rangle$:

$$\rho_{\text{Haar}}(\alpha) = \sum_{N \geqslant 0} \underbrace{e^{-|\alpha|^2} \frac{|\alpha|^{2N}}{N!}}_{} \underbrace{\binom{d+N-1}{N}^{-1} \sum_{n_1+\cdots+n_d=N} |n_1, \ldots, n_d\rangle\langle n_1, \ldots, n_d|}_{}$$

Haar-random coherent states

Average the mapped state over Haar-random $|\psi\rangle$:

$$\rho_{\text{Haar}}(\alpha) = \sum_{N \geqslant 0} \underbrace{e^{-|\alpha|^2} \frac{|\alpha|^{2N}}{N!}}_{\text{Poisson statistics}} \underbrace{\binom{d+N-1}{N}^{-1} \sum_{n_1+\cdots+n_d=N} |n_1, \ldots, n_d\rangle\langle n_1, \ldots, n_d|}$$

Haar-random coherent states

Average the mapped state over Haar-random $|\psi\rangle$:

$$\rho_{\text{Haar}}(\alpha) = \sum_{N \geqslant 0} \underbrace{e^{-|\alpha|^2} \frac{|\alpha|^{2N}}{N!}}_{\text{Poisson statistics}} \underbrace{\binom{d+N-1}{N}^{-1} \sum_{n_1+\cdots+n_d=N} |n_1, \ldots, n_d\rangle\langle n_1, \ldots, n_d|}_{N \text{ photons spread uniformly over } d \text{ modes} \cong \rho_{\mathcal{U}_d^N}}$$

Haar-random coherent states

Average the mapped state over Haar-random $|\psi\rangle$:

$$\rho_{\text{Haar}}(\alpha) = \sum_{N \geqslant 0} \underbrace{e^{-|\alpha|^2} \frac{|\alpha|^{2N}}{N!}}_{\text{Poisson statistics}} \underbrace{\binom{d+N-1}{N}^{-1} \sum_{n_1+\cdots+n_d=N} |n_1, \ldots, n_d\rangle\langle n_1, \ldots, n_d|}_{N \text{ photons spread uniformly over } d \text{ modes} \cong \rho_{\mathcal{U}_d^N}}$$

$\Longrightarrow$ the mean number of photons $|\alpha|^2$ plays the role of the number of copies t

Haar-random coherent states

Average the mapped state over Haar-random $|\psi\rangle$:

$$\rho_{\text{Haar}}(\alpha) = \sum_{N \geqslant 0} \underbrace{e^{-|\alpha|^2} \frac{|\alpha|^{2N}}{N!}}_{\text{Poisson statistics}} \underbrace{\binom{d+N-1}{N}^{-1} \sum_{n_1+\cdots+n_d=N} |n_1,\ldots,n_d\rangle\langle n_1,\ldots,n_d|}_{N \text{ photons spread uniformly over } d \text{ modes} \cong \rho_{\mathcal{U}_d^N}}$$

$\Longrightarrow$ the mean number of photons $|\alpha|^2$ plays the role of the number of copies t

Pseudorandom coherent states (PRCS)

A keyed family of d-mode coherent states $\bigotimes_x |\alpha_{\mathsf{k},x}\rangle$ that is

- normalized: one photon on average,
- efficiently preparable from k,
- pseudorandom: $\bigotimes_x |\sqrt{t}\,\alpha_{\mathsf{k},x}\rangle \approx_c \rho_{\text{Haar}}(\sqrt{t})$ for all $t = \text{poly}(\lambda)$.

A PRCS construction

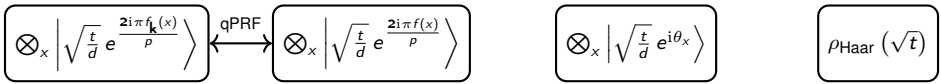
Theorem

If $\{f_k\}_k$ is a quantum-secure PRF from $[2^\lambda]$ to $[p]$, then $\left\{ \bigotimes_{x=0}^{2^\lambda-1} \left| \frac{e^{\frac{2i\pi f_k(x)}{p}}}{\sqrt{2^\lambda}} \right\rangle \right\}_k$ is a PRCS.

A PRCS construction

Theorem

If $\{f_k\}_k$ is a quantum-secure PRF from $[2^\lambda]$ to $[p]$, then $\left\{ \bigotimes_{x=0}^{2^\lambda-1} \left| \frac{e^{\frac{2i\pi f_k(x)}{p}}}{\sqrt{2^\lambda}} \right\rangle \right\}_k$ is a PRCS.



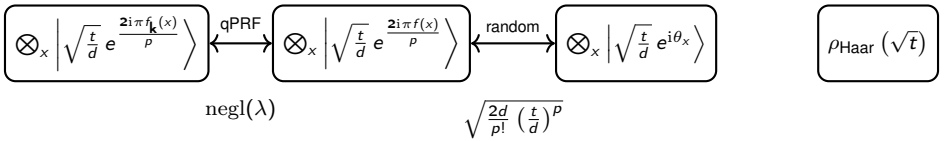
$\text{negl}(\lambda)$

(trace distance bounds up to lower-order terms, with $d = 2^\lambda$)

A PRCS construction

Theorem

If $\{f_k\}_k$ is a quantum-secure PRF from $[2^\lambda]$ to $[p]$, then $\left\{ \bigotimes_{x=0}^{2^\lambda-1} \left| \frac{e^{\frac{2i\pi f_k(x)}{p}}}{\sqrt{2^\lambda}} \right\rangle \right\}_k$ is a PRCS.

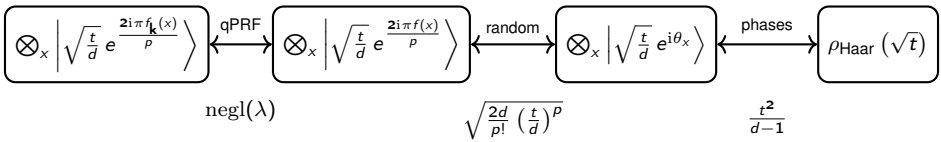


(trace distance bounds up to lower-order terms, with $d = 2^\lambda$)

A PRCS construction

Theorem

If $\{f_k\}_k$ is a quantum-secure PRF from $[2^\lambda]$ to $[p]$, then $\left\{ \bigotimes_{x=0}^{2^\lambda-1} \left| \frac{e^{\frac{2i\pi f_k(x)}{p}}}{\sqrt{2^\lambda}} \right\rangle \right\}_k$ is a PRCS.

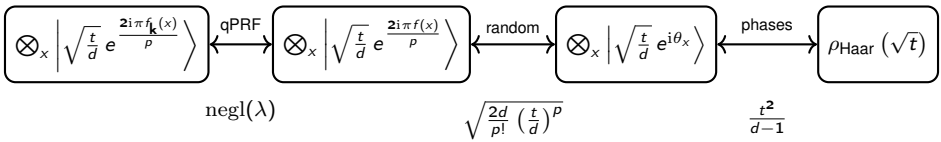


(trace distance bounds up to lower-order terms, with $d = 2^\lambda$)

A PRCS construction

Theorem

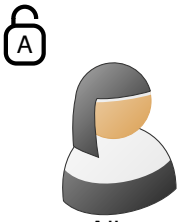
If $\{f_k\}_k$ is a quantum-secure PRF from $[2^\lambda]$ to $[p]$, then $\left\{ \bigotimes_{x=0}^{2^\lambda-1} \left| \frac{e^{\frac{2i\pi f_k(x)}{p}}}{\sqrt{2^\lambda}} \right\rangle \right\}_k$ is a PRCS.



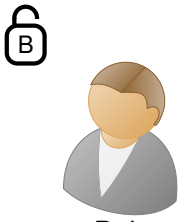
(trace distance bounds up to lower-order terms, with $d = 2^\lambda$)

- Each mode prepared independently: constant depth…
- …but 2^λ modes: practical limitation

An application: cryptography from a riddle for children







Alice

Bob

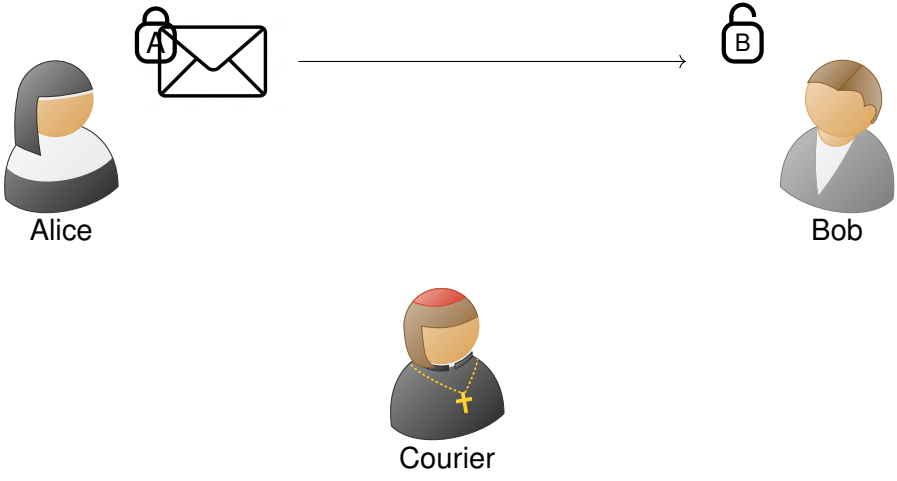


Courier

An application: cryptography from a riddle for children



An application: cryptography from a riddle for children



An application: cryptography from a riddle for children



Alice





Bob



Courier

An application: cryptography from a riddle for children



Alice



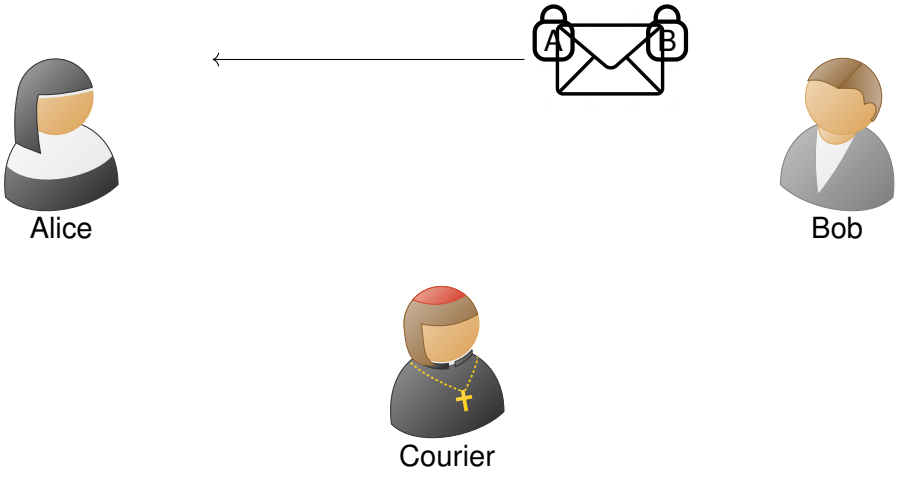


Bob

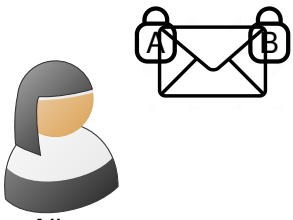


Courier

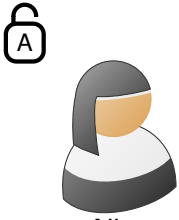
An application: cryptography from a riddle for children



An application: cryptography from a riddle for children



An application: cryptography from a riddle for children





Alice

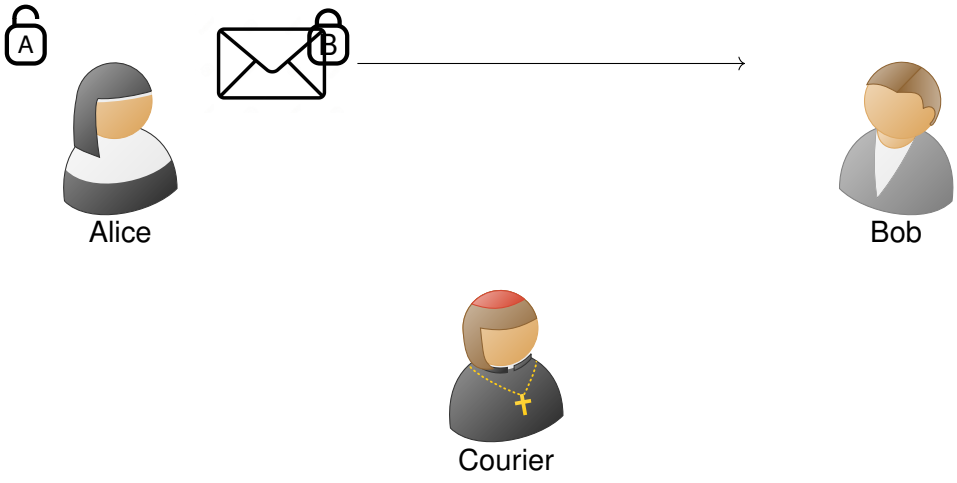


Bob

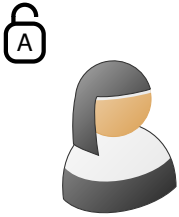


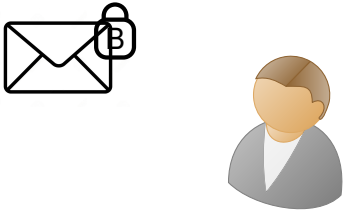
Courier

An application: cryptography from a riddle for children



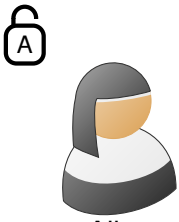
An application: cryptography from a riddle for children





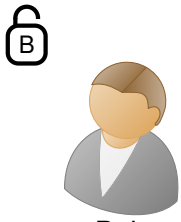


An application: cryptography from a riddle for children



Alice



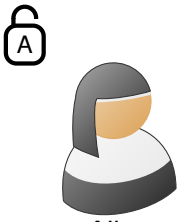


Bob



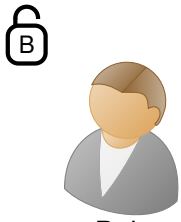
Courier

An application: cryptography from a riddle for children



Alice





Bob



Courier

A message exchanged
without any shared key!

The (coherent) double-lock protocol

Locks must commute. One-time pad? $m \oplus \mathrm{k}_A$, $m \oplus \mathrm{k}_A \oplus \mathrm{k}_B$, $m \oplus \mathrm{k}_B$



Alice, k



Bob, $m \in \{0,1\}^n$

The (coherent) double-lock protocol

Locks must commute. One-time pad? $m \oplus \mathsf{k}_A$, $m \oplus \mathsf{k}_A \oplus \mathsf{k}_B$, $m \oplus \mathsf{k}_B \implies$
XOR them to get m



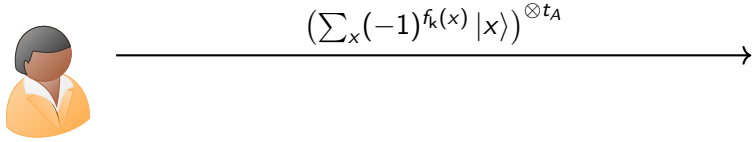
Alice, k



Bob, $m \in \{0,1\}^n$

The (coherent) double-lock protocol

Locks must commute. One-time pad? $m \oplus \mathsf{k}_A$, $m \oplus \mathsf{k}_A \oplus \mathsf{k}_B$, $m \oplus \mathsf{k}_B \implies$
XOR them to get m

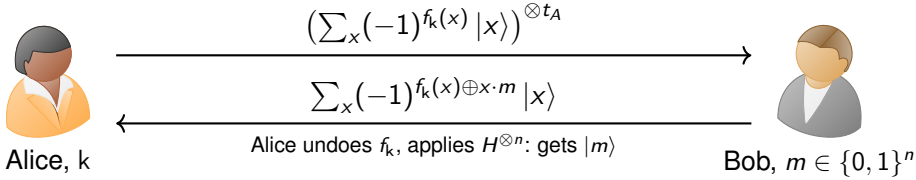


- Alice's state is a PRS, i.e. a *quantum public key*: secure for $t_A = \mathrm{poly}$, $t_B = 1$

  (Coladangelo, 2023; Ananth et al., 2025)

The (coherent) double-lock protocol

Locks must commute. One-time pad? $m \oplus \mathrm{k}_A$, $m \oplus \mathrm{k}_A \oplus \mathrm{k}_B$, $m \oplus \mathrm{k}_B \implies$
XOR them to get m

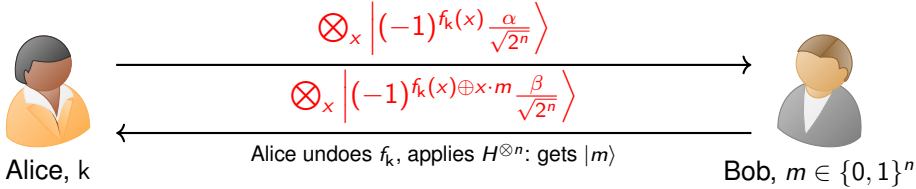


- Alice's state is a PRS, i.e. a *quantum public key*: secure for $t_A = \mathrm{poly}$, $t_B = 1$

  (Coladangelo, 2023; Ananth et al., 2025)

The (coherent) double-lock protocol

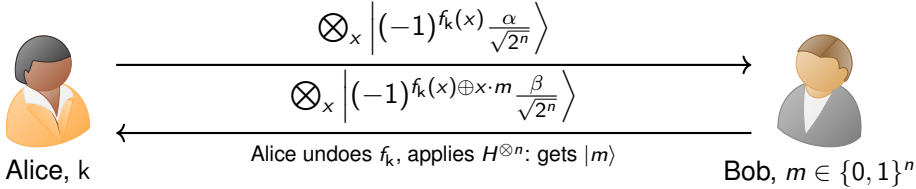
Locks must commute. One-time pad? $m \oplus \mathsf{k}_A$, $m \oplus \mathsf{k}_A \oplus \mathsf{k}_B$, $m \oplus \mathsf{k}_B \implies$ XOR them to get m



- Alice's state is a PRS, i.e. a *quantum public key*: secure for $t_A = \mathrm{poly}$, $t_B = 1$
  (Coladangelo, 2023; Ananth et al., 2025)
- Coherent version: $|\beta| \leqslant |\alpha|$ models losses...

The (coherent) double-lock protocol

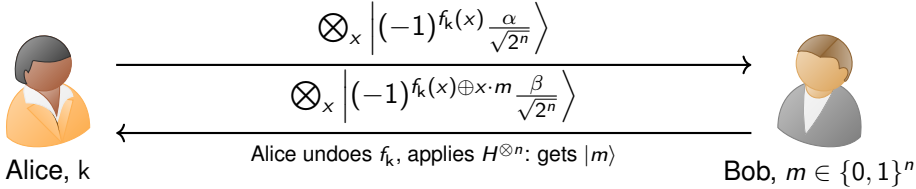
Locks must commute. One-time pad? $m \oplus \mathrm{k}_A$, $m \oplus \mathrm{k}_A \oplus \mathrm{k}_B$, $m \oplus \mathrm{k}_B \implies$
XOR them to get m



- Alice's state is a PRS, i.e. a *quantum public key*: secure for $t_A = \mathrm{poly}$, $t_B = 1$
  (Coladangelo, 2023; Ananth et al., 2025)
- Coherent version: $|\beta| \leqslant |\alpha|$ models losses… and **splitting** the public key among many Bobs

The (coherent) double-lock protocol

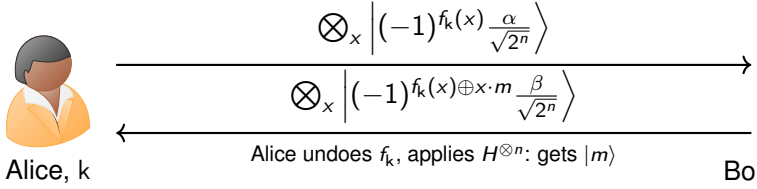
Locks must commute. One-time pad? $m \oplus \mathrm{k}_A$, $m \oplus \mathrm{k}_A \oplus \mathrm{k}_B$, $m \oplus \mathrm{k}_B \implies$ XOR them to get m



- Alice's state is a PRS, i.e. a *quantum public key*: secure for $t_A = \mathrm{poly}$, $t_B = 1$
  (Coladangelo, 2023; Ananth et al., 2025)
- Coherent version: $|\beta| \leqslant |\alpha|$ models losses… and **splitting** the public key among many Bobs
- Public-key functionality from quantum-secure PRFs only, impossible classically
  (Impagliazzo and Rudich, 1990)

The (coherent) double-lock protocol

Locks must commute. One-time pad? $m \oplus \mathrm{k}_A$, $m \oplus \mathrm{k}_A \oplus \mathrm{k}_B$, $m \oplus \mathrm{k}_B \implies$
XOR them to get m



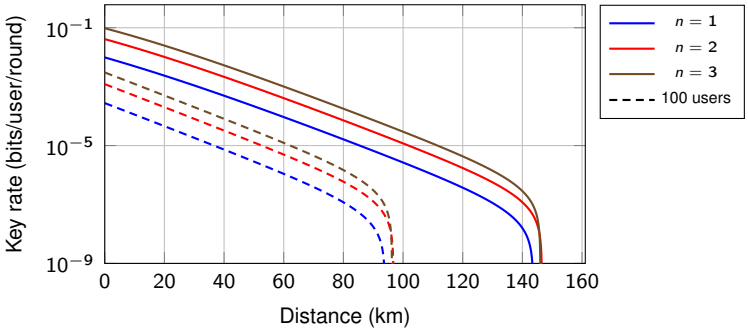
- Alice's state is a PRS, i.e. a *quantum public key*: secure for $t_A = \mathrm{poly}$, $t_B = 1$
  (Coladangelo, 2023; Ananth et al., 2025)
- Coherent version: $|\beta| \leqslant |\alpha|$ models losses… and **splitting** the public key among many Bobs
- Public-key functionality from quantum-secure PRFs only, impossible classically
  (Impagliazzo and Rudich, 1990)
  (assuming an authenticated quantum channel)

Security and key rates

$$p_{\text{guess}} \leqslant 2^{-n/2}\left(\sqrt{e^{2|\alpha||\beta|}+N^2}+N\right)+\Theta\left(2^{-n}N^2\right)$$

Security and key rates

$$p_{\text{guess}} \leqslant 2^{-n/2}\left(\sqrt{e^{2|\alpha||\beta|}+N^2}+N\right)+\Theta\left(2^{-n}N^2\right)$$



Signal split among 100 users: **still non-zero rates** ($T = 10^{-0.02L}$, $\eta_{\text{det}} = 65\%$, $p_{\text{dark}} = 10^{-8}$)

Conclusion

	Contribution	Open problem
QRNG	Maximal intrinsic randomness in closed form, for all sandwiched Rényi entropies and POVMs $\Longrightarrow$ simple benchmark for sources	Discontinuities

Conclusion

	Contribution	**Open problem**
QRNG	Maximal intrinsic randomness in closed form, for all sandwiched Rényi entropies and POVMs $\implies$ simple benchmark for sources	Discontinuities
Real-valued designs	Spectral decomposition of orthogonally-twirled states $\implies \varepsilon = \Omega\left(\frac{t^2}{d}\right)$; imaginarity testing & estimation, real state estimation	Efficient distinguisher $\implies$ real PRS bound

Conclusion

	Contribution	Open problem
QRNG	Maximal intrinsic randomness in closed form, for all sandwiched Rényi entropies and POVMs $\implies$ simple benchmark for sources	Discontinuities
Real-valued designs	Spectral decomposition of orthogonally-twirled states $\implies \varepsilon = \Omega\left(\frac{t^2}{d}\right)$; imaginarity testing & estimation, real state estimation	Efficient distinguisher $\implies$ real PRS bound
PRCS	Haar-random coherent states, PRCS and a construction; quantum public-key encryption with many recipients	Signal strength, authentication, number of modes

Publications and activities

- Tristan Nemoz, Kriss Gutierrez Anco, et al. (2024). *How much secure randomness is in a quantum state?* Submitted to Quantum. arXiv: 2410.16447 [quant-ph]
- Tristan Nemoz, Romain Alléaume, et al. (May 2026). "Exact distinguishability between real-valued and complex-valued Haar random quantum states". In: *Quantum* 10, p. 2120. ISSN: 2521-327X. DOI: 10.22331/q-2026-05-29-2120
- Błażej Kuzaka et al. (In preparation). *Pseudorandom Coherent States: a Practical Path for the Optical Implementation of Uncloneable Cryptography*.
- Tristan Nemoz, Zoé Amblard, et al. (Aug. 2023). "Characterizing the qIND-qCPA (in)security of the CBC, CFB, OFB and CTR modes of operation". In: *Post-Quantum Cryptography*. Ed. by Thomas Johansson and Daniel Smith-Tone. Cham: Springer Nature Switzerland, pp. 445–475. DOI: 10.1007/978-3-031-40003-2_17. URL: https://eprint.iacr.org/2022/236 *(from my master's thesis)*
- Romain Alléaume and Tristan Nemoz (Mar. 2025). "Approches hybrides en cryptographie quantique". In: *Photoniques* 130, pp. 46–48. DOI: 10.1051/photon/202513046 *(popular science)*
- Teaching: Qiskit hackathons for L3 students (exercises, notebooks, 12 hours of lectures)
- Software: `ncpoleon`, (non)commutative polynomial optimization via SDP relaxations (with Peter Brown)

Thank you!

What does the optimal POVM for qubits look like?

For $\rho_A = \lambda_0 |0\rangle\langle 0| + (1-\lambda_0)|1\rangle\langle 1|$ with $\lambda_0 \geqslant \frac{1}{2}$, the rank-1 POVM $\{|\varphi_i\rangle\langle\varphi_i|\}_{i=0}^{3}$ with

$$|\varphi_0\rangle = \frac{1}{2\sqrt{\lambda_0}}\,|0\rangle$$

$$|\varphi_{1+k}\rangle = \sqrt{\frac{4\lambda_0 - 1}{12\lambda_0}}\,|0\rangle + \frac{e^{\frac{2ik\pi}{3}}}{\sqrt{3}}\,|1\rangle \qquad k \in \{0,1,2\}$$

is extremal and yields a uniform distribution when measuring ρ_A.

General dimension: an explicit inductive construction peels off one eigenvalue at a time (thesis, Section 3.4.4).

Is having the POVM rank-1 necessary for independence with the device's inner system?

We proved that $\{M_i\}_i$ rank-1 extremal POVM $\implies \rho_{XE} = f(\rho_{AE}) \perp\!\!\!\perp B$.

Is having the POVM rank-1 necessary for independence with the device's inner system?

Necessary?

We proved that $\{M_i\}_i$ rank-1 extremal POVM $\implies \rho_{XE} = f(\rho_{AE}) \perp\!\!\!\perp B$.

Is having the POVM rank-1 necessary for independence with the device's inner system?

We proved that $\{M_i\}_i$ rank-1 extremal POVM $\implies \rho_{XE} = f(\rho_{AE}) \perp\!\!\!\perp B$.
Dai et al., 2023, Theorem 1: extremality is enough, in a model without A–B entanglement.

Is having the POVM rank-1 necessary for independence with the device's inner system?

We proved that $\{M_i\}_i$ rank-1 extremal POVM $\implies \rho_{XE} = f(\rho_{AE}) \perp\!\!\!\perp B$.
Dai et al., 2023, Theorem 1: extremality is enough, in a model without A–B entanglement.
Generalization removing the rank-1 assumption: Senno, Davino and Mugnai (private communication, unpublished).

Is having the POVM rank-1 necessary for independence with the device's inner system?

We proved that $\{M_i\}_i$ rank-1 extremal POVM $\implies \rho_{XE} = f(\rho_{AE}) \perp\!\!\!\perp B$.
Dai et al., 2023, Theorem 1: extremality is enough, in a model without A–B entanglement.
Generalization removing the rank-1 assumption: Senno, Davino and Mugnai (private communication, unpublished).
Inconsequential for the maximal value: rank-1 extremal POVMs are already optimal.

Where is the intrinsic randomness continuous?

Theorem (rank-1 extremal POVMs)

For a continuous conditional entropy $\mathbb{H}$ and $N \leqslant d_A^2$, $M \mapsto \mathcal{I}_{\mathbb{H}}(\rho_A, M)$ is continuous on rank-1 extremal POVMs with N outcomes: ρ_{XE} only depends on M and a purification of ρ_A.

Theorem (full-rank POVMs)

If $M \mapsto 2^{\mathcal{I}_{\mathbb{H}}(\rho_A, M)}$ is convex (true for sandwiched Rényi entropies), then $M \mapsto \mathcal{I}_{\mathbb{H}}(\rho_A, M)$ is continuous on $\{M \,|\, \forall x, M_x \succ 0\}$, the relative interior

(Rockafellar, 1970).

- Any implemented (noisy) POVM is full-rank...
- ...but can the optimal value be approached inside this set?

Computing the optimal α

Minimize over $\alpha \in (1, 2]$:

$$f(\alpha) = H_{\frac{\alpha}{2\alpha-1}}(A) - \frac{\alpha}{n(\alpha-1)} \log(\varepsilon)$$

- With $p = \mathrm{spec}(\rho_A)$ and $q(\beta)_x \propto p_x^\beta$: $f'(\alpha) = \frac{D_{\mathrm{KL}}\left(q\left(\frac{\alpha}{2\alpha-1}\right)\middle\|p\right)+\frac{\log\varepsilon}{n}}{(\alpha-1)^2}$
- $\alpha \mapsto D_{\mathrm{KL}}\left(q\left(\frac{\alpha}{2\alpha-1}\right)\middle\|p\right)$ is strictly increasing unless p is uniform on its support
- $\implies$ f is unimodal: bisection gives α^* to precision δ in $\left\lceil \log_2 \frac{1}{\delta} \right\rceil - 1$ steps
- As $n \to \infty$: $\alpha = 1 + \frac{1}{\sqrt{n}}$ recovers the asymptotic rate

Eigenvalues of $\rho_{\mathcal{O}_d^t}(\delta)$

For $d = 2$, with T_n the Chebyshev polynomials of the first kind:

$$\lambda_{2,t,k}(\delta) = \frac{t!\,(1-\delta)^{\frac{t}{2}}\; T_{t-2k}\left(\frac{1}{\sqrt{1-\delta}}\right)}{(2t-2k)!!\,(2k)!!}$$

For $d \geqslant 3$, with $C_n^{(\mu)}$ the Gegenbauer polynomials:

$$\lambda_{d,t,k}(\delta) = \frac{t!\,(t-2k)!\,(d-4)!!\,(d-2)!\,(1-\delta)^{\frac{t}{2}}\; C_{t-2k}^{\left(\frac{d}{2}-1\right)}\left(\frac{1}{\sqrt{1-\delta}}\right)}{(2t+d-2k-2)!!\,(2k)!!\,(d+t-2k-3)!}$$

- Proof: Bombieri norm of the projection of P^t onto each $\|X\|^{2k}\,\mathcal{H}_d^{t-2k}$, with $P = \alpha X_0 + \mathrm{i}\beta X_1$
- $\delta = 1$: $\rho_{\mathcal{O}_d^t}(1)$ is maximally mixed on $\mathcal{H}_d^t$

Imaginarity testing

A (β_0, β_1, t)-tester decides whether $\mathsf{I} \leqslant \beta_0$ or $\mathsf{I} \geqslant \beta_1$ from t copies.

Theorem

- If $\beta_1 < \left(\frac{1}{3}\right)^{\frac{2}{d-1}}$, a tester needs $t \sim \sqrt{2d \ln\left(\frac{1}{\frac{2}{3}+\beta_1^{\frac{d-1}{2}}}\right)}$ copies.
- If $\beta_0 > \left(\frac{2}{3}\right)^{\frac{2}{d-1}}$, a tester needs $t \sim d\,\frac{\beta_0^{\frac{d-1}{2}} - \frac{2}{3} + \sqrt{\beta_0^{\frac{d-1}{2}} - \frac{2}{3}}}{\frac{5}{3} - \beta_0^{\frac{d-1}{2}}}$ copies.

Proof: by the exact-design decomposition, the twirl restricted to imaginarities $\geqslant \beta_1$ is $\beta_1^{\frac{d-1}{2}}$-close to $\rho_{\mathcal{U}_d^t}$, so a tester forces $\frac{1}{2}\left\|\rho_{\mathcal{O}_d^t}(0) - \rho_{\mathcal{U}_d^t}\right\|_1 \geqslant \frac{1}{3} - \beta_1^{\frac{d-1}{2}}$; conclude with the asymptotics of the main result.

Key rate ingredients

Alice uses 2^n threshold detectors (efficiency η_{det}, dark counts p_{dark}), transmittance T:

$$1 - p_{\text{abort}} = (1 - p_{\text{dark}})^{2^n - 1} \left[2^n p_{\text{dark}} - \eta_{\text{det}} \left(1 - e^{-T|\beta|^2} \right) (2^n p_{\text{dark}} - 1) \right]$$

$$I(A:B) = n + \Pr[A = B] \log \Pr[A = B] + (1 - \Pr[A = B]) \log \frac{1 - \Pr[A = B]}{2^n - 1}$$

$$p_{\text{guess}} \leqslant \sqrt{\frac{e^{2|\alpha||\beta|} + \left(|\alpha|^2 + |\beta|^2\right)^2}{2^n}} + \frac{|\alpha|^2 + |\beta|^2}{2^{\frac{n}{2}}} + \Theta\left(\frac{\left(|\alpha|^2 + |\beta|^2\right)^2}{2^n} \right)$$

Pessimistic: Eve gets a copy of both the public key and the ciphertext.

Comparison with other key distribution methods

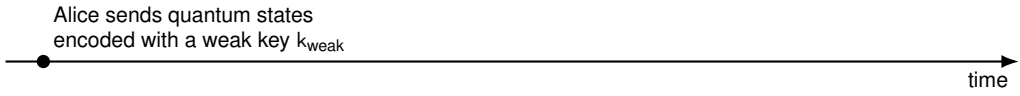
Classical PKE needs *trapdoor* one-way functions (broken by Shor); ours only quantum-secure PRFs, but needs dedicated hardware

Classical SKE needs a pre-shared secret; so do we in practice, to authenticate the quantum channel $\Longrightarrow$ currently the better option

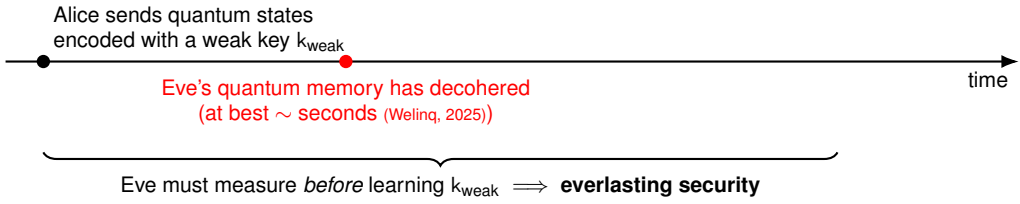
QKD everlasting security, but one key per quantum channel; ours: computational security, one broadcast for many users

Other QPKE qudit-based (Barooti et al., 2023; Coladangelo, 2023; Kitagawa et al., 2024; Malavolta and Walter, 2024); (Kitagawa et al., 2024; Malavolta and Walter, 2024) resist tampered public keys, unlike ours

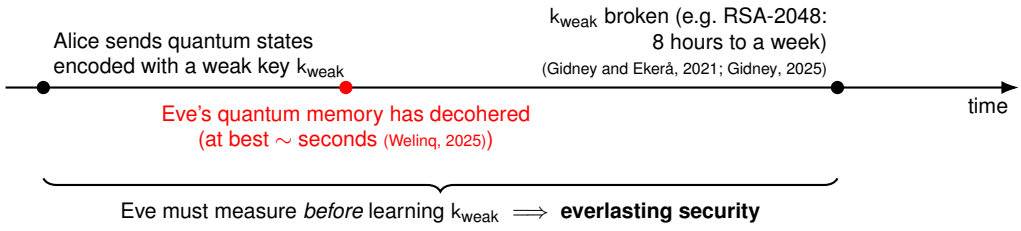
The Quantum Computational Timelock (QCT) model



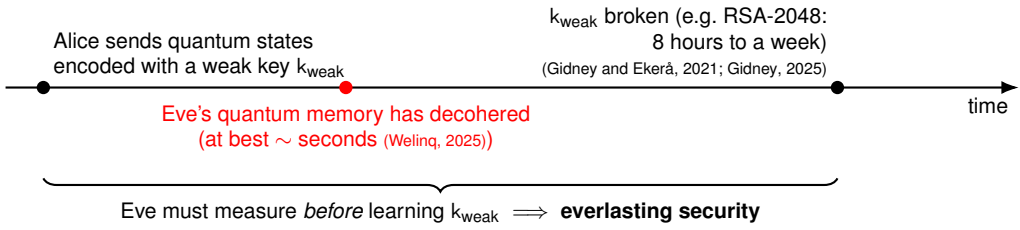
The Quantum Computational Timelock (QCT) model



The Quantum Computational Timelock (QCT) model

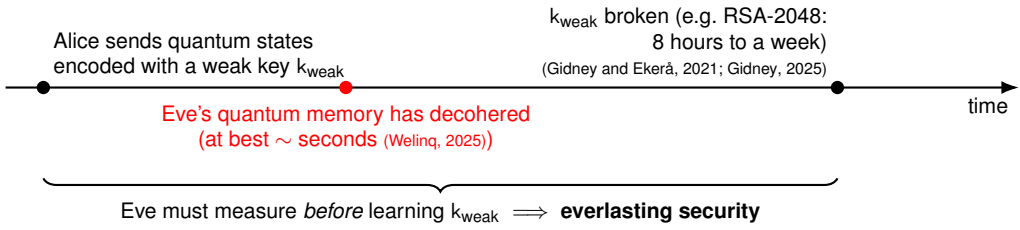


The Quantum Computational Timelock (QCT) model



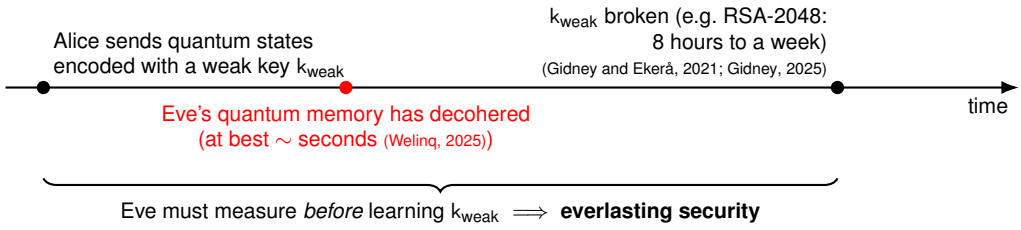
- Mazzoncini et al. (2025): Hidden Matching ($\sqrt{d}$ vs. $\log d$ bits) $\implies$ many copies, but one bit per round and a pessimistic Holevo bound

The Quantum Computational Timelock (QCT) model



- Mazzoncini et al. (2025): Hidden Matching ($\sqrt{d}$ vs. $\log d$ bits) $\implies$ many copies, but one bit per round and a pessimistic Holevo bound
- Stronger separations, e.g. k-Forrelation ($d^{1-\frac{1}{k}}$ vs. $\log d$): a first reduction fails

The Quantum Computational Timelock (QCT) model



- Mazzoncini et al. (2025): Hidden Matching ($\sqrt{d}$ vs. $\log d$ bits) $\implies$ many copies, but one bit per round and a pessimistic Holevo bound
- Stronger separations, e.g. k-Forrelation ($d^{1-\frac{1}{k}}$ vs. $\log d$): a first reduction fails
- More bits per round: double-lock in the QCT model, secure via post-measurement information (Ballester et al., 2008; Gopal and Wehner, 2010)

Forrelation in the QCT model

k-Forrelation: decide whether $\langle 0| HZ_{f_1} H \cdots HZ_{f_k} H |0\rangle$ is close to 0 (Aaronson, 2010; Aaronson and Ambainis, 2015).

- One-way protocol: Alice sends $Z_{f_1} HZ_{g_2} Z_{f_2} H \cdots HZ_{g_k} Z_{f_k} H |0\rangle$ and $\mathrm{Enc}_{k_{\mathrm{weak}}}(g_1)$
- Bob computes the overlap with $Z_{g_1} H |0\rangle$ and decides the secret bit
- Hope: Eve needs $d^{1-\frac{1}{k}}$ classical bits, but each copy gives her only $\log d$ bits

Forrelation in the QCT model

k-Forrelation: decide whether $\langle 0| HZ_{f_1} H \cdots HZ_{f_k} H |0\rangle$ is close to 0 (Aaronson, 2010; Aaronson and Ambainis, 2015).

- One-way protocol: Alice sends $Z_{f_1} HZ_{g_2} Z_{f_2} H \cdots HZ_{g_k} Z_{f_k} H |0\rangle$ and $\mathrm{Enc}_{k_{\mathrm{weak}}}(g_1)$
- Bob computes the overlap with $Z_{g_1} H |0\rangle$ and decides the secret bit
- Hope: Eve needs $d^{1-\frac{1}{k}}$ classical bits, but each copy gives her only $\log d$ bits

Issue (found by Jeanne Lucas): Eve's information is about the *state*, not the functions; once g_1 leaks, computing the overlap may be much easier $\implies$ the reduction does not hold.

Bibliography I

Aaronson, Scott (2010). “BQP and the polynomial hierarchy”. In: *Proceedings of the Forty-Second ACM Symposium on Theory of Computing*. STOC ’10. Cambridge, Massachusetts, USA: Association for Computing Machinery, pp. 141–150. ISBN: 9781450300506. DOI: 10.1145/1806689.1806711. arXiv: 0910.4698 [quant-ph].

Aaronson, Scott and Andris Ambainis (2015). “Forrelation: A Problem that Optimally Separates Quantum from Classical Computing”. In: *Proceedings of the Forty-Seventh Annual ACM Symposium on Theory of Computing*. STOC ’15. Portland, Oregon, USA: Association for Computing Machinery, pp. 307–316. ISBN: 9781450335362. DOI: 10.1145/2746539.2746547. arXiv: 1411.5729 [quant-ph].

Agence Nationale de la Sécurité des Systèmes d’Information (Jan. 25, 2024). *Position Paper on Quantum Key Distribution*. URL: https://cyber.gouv.fr/nous-connaitre/publications/publications-internationales/position-paper-on-quantum-key-distribution/ (visited on 06/02/2026).

Alléaume, Romain and Tristan Nemoz (Mar. 2025). “Approches hybrides en cryptographie quantique”. In: *Photoniques* 130, pp. 46–48. DOI: 10.1051/photon/202513046.

Ananth, Prabhanjan, Aditya Gulati, and Yao-Ting Lin (2025). “Cryptography in the Common Haar State Model: Feasibility Results and Separations”. In: *Theory of Cryptography*. Ed. by Elette Boyle and Mohammad Mahmoody. Cham: Springer Nature Switzerland, pp. 94–125. ISBN: 978-3-031-78017-2. arXiv: 2407.07908 [quant-ph].

Bibliography II

Arrazola, Juan Miguel and Norbert Lütkenhaus (Oct. 2014). “Quantum communication with coherent states and linear optics”. In: *Phys. Rev. A* 90.4, p. 042335. DOI: 10.1103/PhysRevA.90.042335. arXiv: 1406.7189 [quant-ph].

Ballester, Manuel A., Stephanie Wehner, and Andreas Winter (Sept. 2008). “State Discrimination With Post-Measurement Information”. In: *IEEE Transactions on Information Theory* 54.9, pp. 4183–4198. ISSN: 0018-9448. DOI: 10.1109/tit.2008.928276. arXiv: quant-ph/0608014.

Barooti, Khashayar et al. (2023). *Public-Key Encryption with Quantum Keys*. arXiv: 2306.07698 [quant-ph].

Bartusek, James et al. (Aug. 16–20, 2021). “On the Round Complexity of Secure Quantum Computation”. In: *Advances in Cryptology – CRYPTO 2021, Part I*. Ed. by Tal Malkin and Chris Peikert. Vol. 12825. Lecture Notes in Computer Science. Virtual Event: Springer, Cham, Switzerland, pp. 406–435. DOI: 10.1007/978-3-030-84242-0_15.

Bennett, Charles H. and Gilles Brassard (Dec. 2014). “Quantum cryptography: Public key distribution and coin tossing”. In: *Theoretical Computer Science* 560, pp. 7–11. ISSN: 0304-3975. DOI: 10.1016/j.tcs.2014.05.025. arXiv: 2003.06557 [quant-ph].

Beullens, Ward (Aug. 2022). “Breaking Rainbow Takes a Weekend on a Laptop”. In: *Advances in Cryptology – CRYPTO 2022, Part II*. Ed. by Yevgeniy Dodis and Thomas Shrimpton. Vol. 13508. Lecture Notes in Computer Science. Santa Barbara, CA, USA: Springer, Cham, Switzerland, pp. 464–479. DOI: 10.1007/978-3-031-15979-4_16. URL: https://ia.cr/2022/214.

Bibliography III


Brakerski, Zvika and Omri Shmueli (Dec. 1–5, 2019). “(Pseudo) Random Quantum States with Binary Phase”. In: *TCC 2019: 17th Theory of Cryptography Conference, Part I*. Ed. by Dennis Hofheinz and Alon Rosen. Vol. 11891. Lecture Notes in Computer Science. Nuremberg, Germany: Springer, Cham, Switzerland, pp. 229–250. DOI: 10.1007/978-3-030-36030-6_10. arXiv: 1906.10611 [quant-ph].


— (Aug. 17–21, 2020). “Scalable Pseudorandom Quantum States”. In: *Advances in Cryptology – CRYPTO 2020, Part II*. Ed. by Daniele Micciancio and Thomas Ristenpart. Vol. 12171. Lecture Notes in Computer Science. Santa Barbara, CA, USA: Springer, Cham, Switzerland, pp. 417–440. DOI: 10.1007/978-3-030-56880-1_15. arXiv: 2004.01976 [quant-ph].


Burchardt, Adam et al. (2025). *High-dimensional quantum Schur transforms*. arXiv: 2509.22640 [quant-ph].


Castryck, Wouter and Thomas Decru (Apr. 2023). “An Efficient Key Recovery Attack on SIDH”. In: *Advances in Cryptology – EUROCRYPT 2023, Part V*. Ed. by Carmit Hazay and Martijn Stam. Vol. 14008. Lecture Notes in Computer Science. Lyon, France: Springer, Cham, Switzerland, pp. 423–447. DOI: 10.1007/978-3-031-30589-4_15. URL: https://ia.cr/2022/975.


Center, National Cyber Security (Mar. 24, 2020). *Quantum security technologies*. URL: https://www.ncsc.gov.uk/paper/quantum-security-technologies (visited on 06/02/2026).

Bibliography IV

Coifman, Ronald Raphaël and Guido Leopold Weiss (1968). “Representations of compact groups and spherical harmonics”. English. In: *L'Enseignement Mathématique. 2e Série* 14, pp. 121–173. ISSN: 0013-8584. DOI: 10.5169/seals-42346.

Coladangelo, Andrea (2023). *Quantum trapdoor functions from classical one-way functions*. arXiv: 2302.12821 [quant-ph].

Cui, Laura et al. (2025). *Unitary designs in nearly optimal depth*. arXiv: 2507.06216 [quant-ph].

Dai, Hao et al. (Aug. 2023). “Intrinsic randomness under general quantum measurements”. In: *Phys. Rev. Res.* 5.3, p. 033081. DOI: 10.1103/PhysRevResearch.5.033081. arXiv: 2203.08624 [quant-ph].

Dupuis, Frédéric (2023). “Privacy Amplification and Decoupling Without Smoothing”. In: *IEEE Transactions on Information Theory* 69.12, pp. 7784–7792. DOI: 10.1109/TIT.2023.3301812. arXiv: 2105.05342 [quant-ph].

Gidney, Craig (2025). *How to factor 2048 bit RSA integers with less than a million noisy qubits*. arXiv: 2505.15917 [quant-ph].

Gidney, Craig and Martin Ekerå (Apr. 2021). “How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits”. In: *Quantum* 5, p. 433. ISSN: 2521-327X. DOI: 10.22331/q-2021-04-15-433.

Gopal, Deepthi and Stephanie Wehner (Aug. 2010). “Using postmeasurement information in state discrimination”. In: *Physical Review A* 82.2. ISSN: 1094-1622. DOI: 10.1103/physreva.82.022326. arXiv: 1003.0716 [quant-ph].

Bibliography V


Grilo, Alex B. et al. (Oct. 17–21, 2021). “Oblivious Transfer is in MiniQCrypt”. In: *Advances in Cryptology – EUROCRYPT 2021, Part II*. Ed. by Anne Canteaut and François-Xavier Standaert. Vol. 12697. Lecture Notes in Computer Science. Zagreb, Croatia: Springer, Cham, Switzerland, pp. 531–561. DOI: 10.1007/978-3-030-77886-6_18.


Haug, Tobias, Kishor Bharti, and Dax Enshan Koh (June 2025). “Pseudorandom unitaries are neither real nor sparse nor noise-robust”. In: *Quantum* 9, p. 1759. ISSN: 2521-327X. DOI: 10.22331/q-2025-06-04-1759. arXiv: 2306.11677 [quant-ph].


Impagliazzo, Russell (1995). “A personal view of average-case complexity”. In: *Proceedings of Structure in Complexity Theory. Tenth Annual IEEE Conference*, pp. 134–147. DOI: 10.1109/SCT.1995.514853. URL: https://cseweb.ucsd.edu/~russell/average.ps.


Impagliazzo, Russell and Steven Rudich (Aug. 1990). “Limits on the Provable Consequences of One-way Permutations”. In: *Advances in Cryptology – CRYPTO’88*. Ed. by Shafi Goldwasser. Vol. 403. Lecture Notes in Computer Science. Santa Barbara, CA, USA: Springer, Heidelberg, Germany, pp. 8–26. DOI: 10.1007/0-387-34799-2_2.


Jeronimo, Fernando Granha, Nir Magrafta, and Pei Wu (2024). *Pseudorandom and Pseudoentangled States from Subset States*. arXiv: 2312.15285 [quant-ph].

Bibliography VI


Ji, Zhengfeng, Yi-Kai Liu, and Fang Song (Aug. 19–23, 2018). “Pseudorandom Quantum States”. In: *Advances in Cryptology – CRYPTO 2018, Part III*. Ed. by Hovav Shacham and Alexandra Boldyreva. Vol. 10993. Lecture Notes in Computer Science. Santa Barbara, CA, USA: Springer, Cham, Switzerland, pp. 126–152. DOI: 10.1007/978-3-319-96878-0_5. URL: https://ia.cr/2018/544.


Kitagawa, Fuyuki et al. (2024). “Quantum Public-Key Encryption with Tamper-Resilient Public Keys from One-Way Functions”. In: *Advances in Cryptology – CRYPTO 2024*. Ed. by Leonid Reyzin and Douglas Stebila. Cham: Springer Nature Switzerland, pp. 93–125. ISBN: 978-3-031-68394-7. DOI: 10.1007/978-3-031-68394-7_4. arXiv: 2304.01800 [quant-ph].


Kretschmer, William (2021). “Quantum Pseudorandomness and Classical Complexity”. In: *16th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2021)*. Ed. by Min-Hsiu Hsieh. Vol. 197. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2:1–2:20. ISBN: 978-3-95977-198-6. DOI: 10.4230/LIPIcs.TQC.2021.2.


Lu, Chuhan et al. (Dec. 2–6, 2024). “Quantum Pseudorandom Scramblers”. In: *TCC 2024: 22nd Theory of Cryptography Conference, Part II*. Ed. by Elette Boyle and Mohammad Mahmoody. Vol. 15365. Lecture Notes in Computer Science. Milan, Italy: Springer, Cham, Switzerland, pp. 3–35. DOI: 10.1007/978-3-031-78017-2_1. arXiv: 2309.08941 [quant-ph].

Bibliography VII


Malavolta, Giulio and Michael Walter (Aug. 2024). “Robust Quantum Public-Key Encryption with Applications to Quantum Key Distribution”. In: *Advances in Cryptology – CRYPTO 2024, Part VII*. Ed. by Leonid Reyzin and Douglas Stebila. Vol. 14926. Lecture Notes in Computer Science. Santa Barbara, CA, USA: Springer, Cham, Switzerland, pp. 126–151. DOI: 10.1007/978-3-031-68394-7_5. arXiv: 2304.02999 [quant-ph].


Mazzoncini, Francesco et al. (Sept. 2025). “Hybrid Quantum Cryptography from Communication Complexity”. In: *Quantum* 9, p. 1862. ISSN: 2521-327X. DOI: 10.22331/q-2025-09-24-1862.


Nemoz, Tristan, Romain Alléaume, and Peter Brown (May 2026). “Exact distinguishability between real-valued and complex-valued Haar random quantum states”. In: *Quantum* 10, p. 2120. ISSN: 2521-327X. DOI: 10.22331/q-2026-05-29-2120.


Nemoz, Tristan, Kriss Gutierrez Anco, and Peter Brown (2024). *How much secure randomness is in a quantum state?* Submitted to Quantum. arXiv: 2410.16447 [quant-ph].


Rockafellar, Ralph Tyrell (1970). *Convex Analysis*. Vol. 28. Princeton Landmarks in Mathematics and Physics. Princeton University Press, pp. 82–92. ISBN: 9780691015866.


Shor, Peter W. (Oct. 1997). “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer”. In: *SIAM Journal on Computing* 26.5, pp. 1484–1509. ISSN: 1095-7111. DOI: 10.1137/s0097539795293172. arXiv: quant-ph/9508027.

Bibliography VIII



Welinq (Mar. 2025). *Welinq Launches Its World-Record Storage Solution for Quantum Computing Scale-Out. A critical step in building the infrastructure for scalable quantum data centers.* URL: https://www.welinq.fr/blog/welinq-launches-its-world-record-storage-solution-for-quantum-computing-scale-out (visited on 04/20/2026).



Wiesner, Stephen (Jan. 1983). "Conjugate coding". In: *SIGACT News* 15.1, pp. 78–88. ISSN: 0163-5700. DOI: 10.1145/1008908.1008920.